

Ericsson Private 5G TCC Factory Solution Guideline



Copyright

© Ericsson AB 2023. All rights reserved. No part of this document may be reproduced in any form without the written permission of the copyright owner.

Disclaimer

The contents of this document are subject to revision without notice due to continued progress in methodology, design, and manufacturing. Ericsson shall have no liability for any error or damage of any kind resulting from the use of this document.

Trademark List

All trademarks mentioned herein are the property of their respective owners. These are shown in the document Trademark Information.



Contents

1	Introduction	4
1.1	General.....	4
1.2	Scope	4
2	Principles and Guidelines	5
2.1	Ericsson Private 5G Solution Overview.....	5
2.1.1	Network Controllers.....	5
2.1.2	Basebands and Radio Equipment	6
2.1.3	Router.....	6
2.1.4	Cloud Management	6
2.1.5	Network Management Portal.....	6
2.1.6	Performance Monitoring	7
2.1.7	API Integration	7
2.2	Application Requirements	7
2.3	Dimensioning	8
2.3.1	Considerations for TCC Industrial Networks.....	8
2.3.2	Network Configuration and Baseband Processing Limitations.....	8
2.3.3	Dimensioning Methodology for Time Critical Low Bitrate Service.....	16
2.4	Radio Network Planning and Deployment Considerations.....	46
2.4.1	SINR considerations.....	46
2.4.2	Radio Considerations for Deployments with Multiple Cells	46
2.5	QoS Considerations	48
3	Network Elements and RAN Features List	50
3.1	RAN Feature List	50
3.2	Radio and Transport Hardware.....	52
4	Network Design – Factory Example	53
4.1	Example 1 – Factory 1 with PROFINET Connections	53
4.1.1	Overview	53
4.1.2	Service Requirements	53
4.1.3	Physical Site Design.....	54
4.1.4	Transport Design	54
4.1.5	Radio Network Design.....	54
4.1.6	UE / Device.....	58
4.1.7	Observability	59
4.1.8	Positioning.....	59
4.2	Example 2 – Factory 2 with PROFINET Connections	59
4.2.1	Overview	59
4.2.2	Service Requirements	59
4.2.3	Radio Network Design.....	59
5	Observability	63
6	Troubleshooting & Best Practices.....	63
7	Solution Guideline Change History.....	64



1 Introduction

The purpose of this document is to support solution architects and integration engineers in planning, design, and deployment of Ericsson Private 5G solution considering introduction of latency sensitive services in industrial environments.

Document present design overview and configuration guidance together with relevant requirements for time-critical deployment.

This document is for internal use only and are therefore only available through CALStore in Network Solutions 23.Q4.0 Library.

1.1 General

This document provides dimensioning and connectivity guidelines for the deployment of Ericsson Private 5G within the context of industrial deployment of Time Critical Services. This guideline supports high-level and low-level design activities by network deployment examples.

For more information on Ericsson Private 5G solution please see reference material in CALStore and Portfolio and Commerce.

- [Ericsson Private 5G 23.71, Internal CALStore](#)
- [Ericsson Private 5G Portfolio and Commerce](#)

For more information on solution design for Indoor and Time Critical Services, Critical-IoT, see the following guidelines and sources in [Network Solutions CALStore](#).

- [NR Indoor and Venue Solution Guideline](#)
- [Critical IoT Solution Guideline](#)
- [E2E QoS Guidelines](#)
- [NR Transport Dimensioning Guidelines](#)
- [NR Network Synchronization Solution Guidelines](#)
- [Ericsson Indoor Planner](#)
- [5G Indoor Positioning Deployment Guideline](#)

1.2 Scope

The following items are within the scope of this document:

- Dimensioning and feature activation recommendations provided in this first release will be limited to cover Low Bitrate services.
- Design examples of Private 5G TCC Factory Network with Radio Dot System deployment recommendations.
- Bands: FR1
- TDD



2 Principles and Guidelines

Scope of this document is to describe the main configurational aspects and recommendations needed for factory indoor deployments, this with reference to corresponding products such as Indoor Radio DoT. As to give a more complete overview of Ericsson Private 5G product portfolio, the following chapter also references the availability of Micro & Macro Radio for larger deployments covering both indoor and outdoor scenarios.

2.1 Ericsson Private 5G Solution Overview

Ericsson Private 5G is a product that consists of software and services that were developed specifically for Ericsson Private 5G, and of products that were developed within Ericsson. A dual core with support for 5G and 4G, running locally on premises with high availability and low latency, is built into Ericsson Private 5G. All network traffic stays on premises, and there are no charges per device or gigabyte of data.

Ericsson Private 5G is end-to-end tested, fully documented, and complemented by extensive training material.

Ericsson Private 5G includes the following main components.

2.1.1 Network Controllers

The network controllers are compact, lifecycle-managed networking appliances that are implemented as a pair of redundant servers, enabling high-availability capabilities. They run all the on-premises software components of Ericsson Private 5G except for software components that run on the basebands, router, and radio equipment.



2.1.2 Basebands and Radio Equipment

Ericsson Private 5G leverages the Ericsson radio portfolio, enabling use cases ranging from compact small-area indoor use cases to wide-area outdoor use cases:

- Radio dots provide indoor coverage and capacity while delivering a compact form factor suited for small area use cases.
- Micro radios with more power, offering wider coverage than radio dots, can be used for both indoor and outdoor use cases.
- Macro radios offer high power and provide outdoor coverage in wide area use cases.
- Antenna integrated radios offer high power and capacity, offer very flexible coverage, and are used for both indoor and outdoor deployments.
- The compute capability for the radio network is provided by Ericsson basebands, RAN processors, and radio processors (all referred to as basebands in this document), featuring custom Ericsson silicon.

2.1.3 Router

Sites that include three or more basebands must use a router for the connections between network controllers and basebands. For sites with only one or two basebands, the basebands can be directly connected to the network controllers. Using a router is also supported. Ericsson Private 5G supports Router 6675 from the Ericsson transport portfolio as a managed component. Routers or switches from other manufacturers are not presently supported by Ericsson Private 5G.

2.1.4 Cloud Management

Centralized cloud-management capabilities are provided by Ericsson to all partners and enterprise customers as part of Ericsson Private 5G. The cloud management capabilities enable remote control and monitoring of sites and lifecycle management such as software updates and upgrades to the site equipment.

2.1.5 Network Management Portal

The main management tool for Ericsson Private 5G is the Network Management Portal.



2.1.6 Performance Monitoring

Ericsson Private 5G provides Service Level Agreement (SLA) monitoring and Key Performance Indicator (KPI) tools. The performance of a site will be verified at installation and handover. The Watchdog app, available for Android-based devices and as headless software running on a custom industrial gateway, referred to as the industrial Watchdog device, allows constant monitoring and verification of the system network performance. The performance monitoring is displayed on the screen of Android based devices and in the Network Management Portal

2.1.7 API Integration

Network management functions for Ericsson Private 5G are implemented through a set of local and cloud Application Programming Interfaces (APIs), developed with RESTful design principles, that provide programmatic access to a subset of capabilities implemented in the Network Management Portal.

The Cloud API enables the integration of customized cloud applications with Ericsson Private 5G and is available to enterprises, CSPs and third-party developers.

Cloud API access to a site is granted by the administrator of the site using an API-key system and encrypted access managed through the Network Management Portal. For more information, see document NMP User Guide - Site Admin and Security Admin.

One key API enables external bulk provisioning of SIM cards.

2.2 Application Requirements

There are different types of applications in a factory environment, each with its own requirements. For example, robots using the PROFINET protocol have strict latency requirements but low uplink bit rates, while applications such as video streaming or augmented reality may have more relaxed latency requirements but demand high uplink bit rates.

The focus of this guideline is on critical low-bit rate applications (e.g., PROFINET). These applications are characterized by:

- Small data packet
- Defined periodicity.

For this type of applications to design and dimension the network, the following information must be collected from the customer:

- Periodicity of packets
- Packet size
- Latency requirement
- Targeted number of devices



- Application behavior upon loss of packets
- Number of allowed lost packets.

2.3 Dimensioning

The main objective of this section is to explain how time critical low-bit rate application requirements such as latency, traffic periodicity, packet size or number of devices can be achieved taking into consideration network limitations like TDD pattern, number of scheduling elements per TTI, bandwidth, etc.

2.3.1 Considerations for TCC Industrial Networks

Two main considerations apply to this TCC dimensioning guideline:

- 1 Good radio coverage is assumed in the factory premises.
- 2 Dimensioning for TCC applications needs to be done assuming the worst-case scenario.

The reason for the first consideration is that contrary to public wide area networks, in which zones with low or no radio coverage are common, in a dedicated local network it is possible for radio planners to design the radio network in such a way that a minimum level of coverage is guaranteed in all areas where users are expected to be present. Therefore, in this guideline it is assumed that appropriate radio coverage or signal strength exists in the factory. In Section 2.4, a list of recommendations is for radio planning in factory environments is given.

The second consideration is that in the case of mobile broadband (MBB) applications, achieving an average throughput or average latency is often sufficient to fulfill the application requirements. However, for TCC applications there are absolute maximum latency and minimum reliability requirements that cannot be addressed by average estimations. Thus, the dimensioning should be done in such a way that the TCC application latency and reliability requirements are fulfilled even in the worst-case scenario.

2.3.2 Network Configuration and Baseband Processing Limitations

The following is a list of network and hardware limitations that should be considered:

- Latency
- TDD pattern
- Uplink scheduling configuration
- Number of supported SE/TTI
- Number of supported CG/TTI
- Carrier bandwidth
- Number of cells and distribution of devices across cells



2.3.2.1 Latency

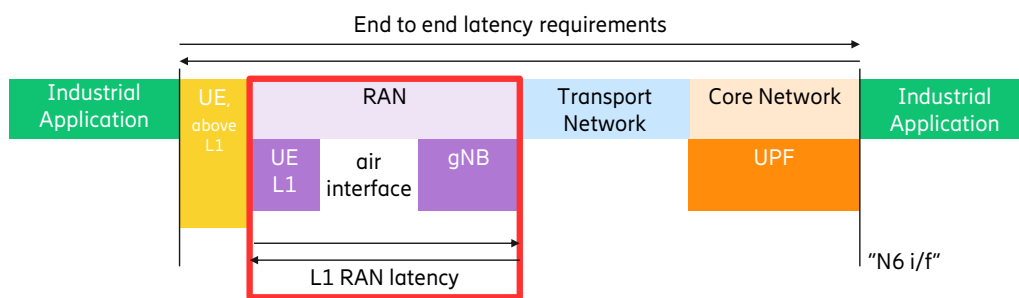


Figure 1 End to end and layer 1 RAN latency

As shown in Figure 1, latency is the time that it takes (including any queuing delay and retransmissions) to prepare and transmit a message successfully across the network. The total end-to-end latency is composed of the delays caused by the core network, the transport network, the radio access network (RAN) and even the UE.

This guideline will focus on layer 1 RAN latency. Therefore, only the layer 1 processing delays in the UE and gNB will be considered.

There are protocols with round trip time requirements (i.e., feedback is required in certain amount of time). In those cases, the downlink must also be taken into consideration in the delay budget. However, in this guideline the focus will be only in the reduction of layer 1 latency in the uplink.

As shown in Figure 1, there are additional delays on top of the layer 1 RAN delay that will not be considered in this guideline but that must be taken into consideration for end-to-end latency calculations.

2.3.2.2 TDD pattern

In NR, a slot can be either used to schedule an uplink or a downlink transmission. A TDD pattern or TDD frame structure is a predefined series of uplink and downlink slots.

Most of the currently used TDD patterns are downlink heavy patterns (e.g., DDDSU) in which the proportion of downlink slots is higher than uplink slots. These patterns are suited for mobile broadband (MBB) applications but may not meet latency and uplink capacity requirements of time critical applications.

For time critical applications services, a balanced TDD pattern (e.g., DDDS1UUDS2UU) with a more equal proportion of uplink and downlink slots helps to increase the number of supported users as more uplink slots are available for transmission.

The selection of the TDD pattern is limited by the TDD patterns used in neighbor networks using the same or adjacent spectrum. Interference occurs when an uplink transmission is scheduled in a dedicated network while a downlink transmission is scheduled in the same instant in a neighbor network. Therefore,



commonly neighbor networks in adjacent frequencies share the same TDD pattern.

For a list of the available TDD patterns in Ericsson RAN software, see [NR RAN CPI documentation](#).

This guideline will focus on the following TDD patterns:

- TDD_ULDL_PATTERN_02: DDDSU, with special slot S=10:2:2
- TDD_ULDL_PATTERN_07: DDDS1UUDS2UU, with special slots S1=6:4:4 and S2=10:4:0

The DDDSU pattern was selected because it is a TDD pattern commonly used by operators and can also be used to provide low RAN latency. The DDDS1UUDS2UU pattern was chosen due to its high uplink capacity (i.e., more users can be scheduled in the uplink in a single frame).

2.3.2.3 Uplink Scheduling Configuration

Both downlink and uplink scheduling are controlled in the gNB. While downlink scheduling can start immediately after downlink data is available in the gNB, uplink scheduling can be more complicated.

In dynamic uplink scheduling, if uplink data is available in the UE and uplink resources are not yet assigned to it, the UE needs to indicate to the gNB the need of resources via a scheduling request message. The UE must wait for a transmission grant message from the gNB to know which uplink resources are allocated for transmission.

In the case of time critical applications with periodic uplink transmission, this extra delay in the uplink is not negligible. Therefore, special attention is required in the selected uplink scheduling method. In this section, three available methods for uplink scheduling are described:

- Dynamic grant scheduling
- Uplink configured grant
- Prescheduling

2.3.2.3.1 Dynamic Grant Scheduling

If dynamic grant scheduling is used for uplink scheduling, the UE sends a scheduling request (SR) to ask for a grant for uplink transmission as shown in Figure 2. Uplink transmission will only start after the gNB process the scheduling request, encodes the uplink grant, and sends it to the UE.

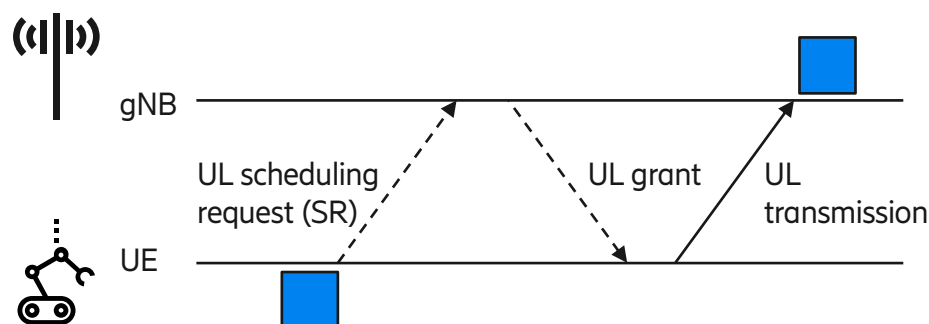


Figure 2 Dynamic grant scheduling

However, if the uplink grant does not allocate enough resources for the uplink transmission the packet will be segmented, and the UE will be required to send a buffer status report (BSR) to request more uplink resources to the gNB. This will increase the transmission delay and the number of PDCCH and PUSCH transmissions.

Therefore, to avoid segmentation and ensure that the packet is transmitted as shown in Figure 2, the initial grant size must be adjusted to match the packet size. This can be achieved with the `srGrantSize` MoM parameters using the [NR Service-Adaptive Link Adaptation](#) feature described in Section 2.3.3.1.5.1.

2.3.2.3.2 Prescheduling

To decrease the latency, the [NR Service-Adaptive Prescheduling](#) feature can be used to reduce the time that the UE has to wait to receive an uplink grant. Instead of waiting for a scheduling request from the UE, the gNB periodically sends uplink grants to the UE without receiving scheduling request or buffer status reports from the UE.

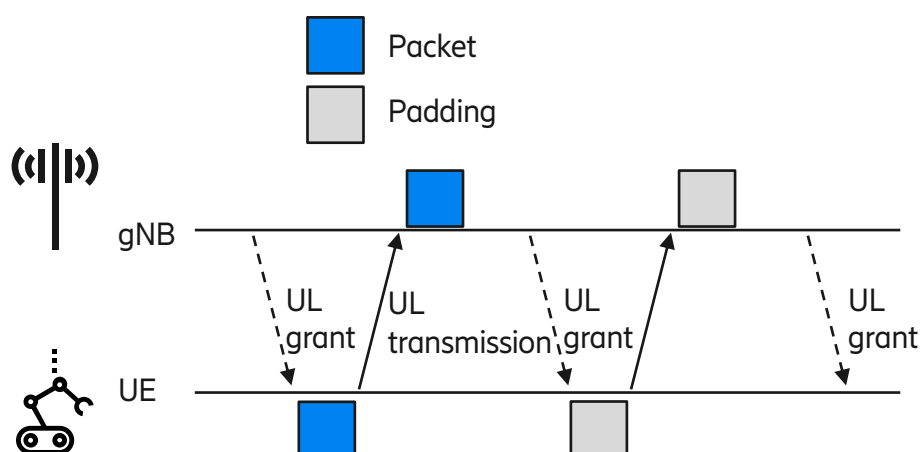


Figure 3 Prescheduling

In Figure 3, it can be seen how the gNB periodically sends uplink grants to the UE to reduce the time that the UE must wait to send the packet. However, it can also be seen that the UE must always transmit upon the reception of an uplink grant. If the UE has not data to transmit, padding is transmitted instead.

Similarly to dynamic grant, it is also important to configure the `preschedulingDataSize` parameter so that the packet is sent in a single to transmission and no segmentation occurs.

As shown in Figure 3, the gNB is periodically sending uplink grants to the UEs. This increases the load of the PDCCH and can lead to PDCCH blocking. Moreover, because of the UE padding transmission, PUSCH interference, and UE power consumption increase are expected.

In scenarios with mixed services and high load, the behavior of UEs with basic pre-scheduling priority (Section 2.3.3.2) is unpredictable as pre-scheduled UEs will get lower priority compared to dynamic grant scheduled UEs.

2.3.2.3.3 Uplink Configured Grant

If uplink configured grant is used instead, the UE is not required to send an SR and wait for an uplink grant to start an uplink transmission. Configured-grant scheduling will significantly improve latency for periodic, deterministic applications since the SR procedure is not needed.

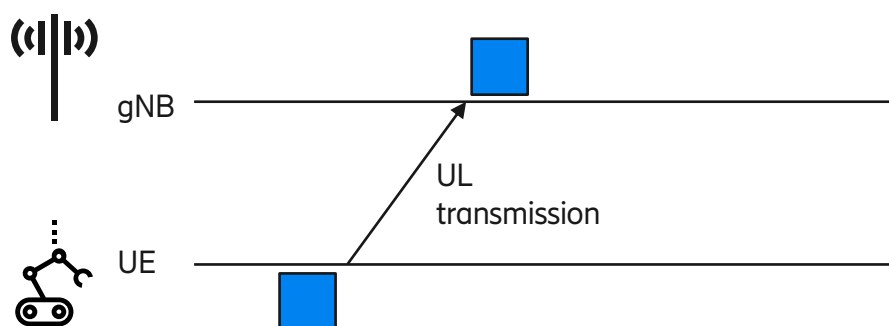


Figure 4 Uplink configured grant scheduling

In the [Uplink Configured Grant for URLLC Dedicated RAN](#) feature, the maximum number of UEs that can transmit during an uplink slot depends on the chosen grant periodicity (Section 2.3.3.3.1) and the maximum number of configured grants per TTI (CG/TTI) supported (Section 2.3.3.3.2).

To avoid segmentation and ensure that the packet is sent in a single transmission as shown in Figure 4, the `cgDataSize MoM` parameter (Section 2.3.3.3.4) has to be configured to match the packet size so that the data can be sent in a single transmission.

Additionally, the Connected Mode DRX feature must be disabled for the group of UEs with low-latency services.

See the [Critical IoT Solution Guideline](#) for more information on how the uplink configured grant is applied in public networks for real-time media use cases (e.g., cloud assisted AR, cloud VR and cloud gaming).

2.3.2.3.4 Comparison of Uplink Scheduling Methods

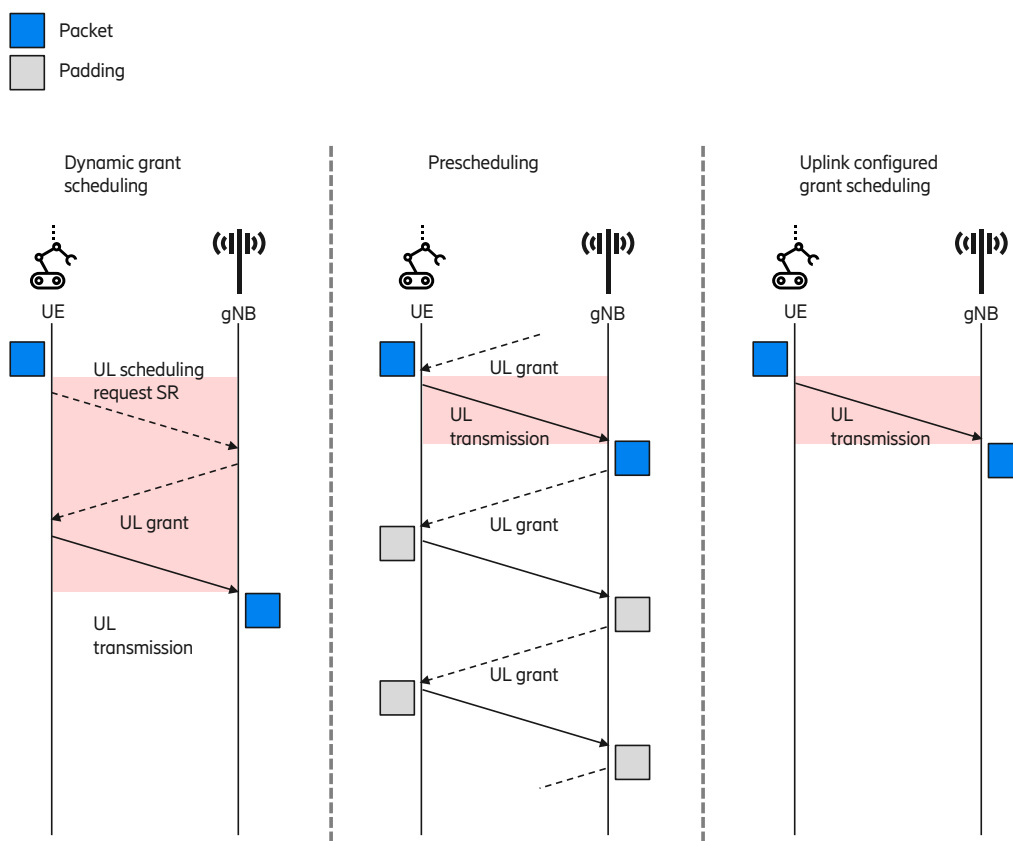


Figure 5 Comparison of different uplink scheduling methods

In Figure 5, a comparison of the 3 scheduling methods presented on this section is shown. The selection of the appropriate method depends on latency requirements, number of supported UEs and UE capabilities:

- Dynamic uplink grant is preferred when many devices must be supported by a single cell and the one way RAN latency requirement is above 7 ms.
- Both prescheduling and uplink configured grant are options for one way RAN ultra-low latency requirements (e.g., under 7 ms).
- Prescheduling is only recommended in cases with low traffic load and where a mix of low and non-low latency services is not expected.
 - Basic priority UEs using prescheduling have lower priority than dynamic grant UEs. In case of congestion, basic priority UEs using prescheduling will be down prioritized.
 - Due to the periodic uplink grants, prescheduling can cause PDCCH blocking and imposes a larger burden on PDCCH resources than uplink configured grant.
 - Because of padding transmission, prescheduling causes PUSCH interference and UE power consumption to increase.
- Uplink configured grant can be used for services requiring ultra-low latency requirements (e.g., under 7 ms) and without stressing the PDCCH.
 - It requires UE support (i.e., configuredUIGrantType2 in Rel. 15 and enhancedSkipUplinkTx in Rel 16).



- Dynamic link adaptation is currently not supported and thus a fixed modulation and coding scheme is used. MCS0 is the default modulation and coding scheme.
- Packet size including headers must not be larger than 10000 bytes (See [Uplink Configured Grant for URLLC Dedicated RAN](#)).

2.3.2.4 Number of Supported SE/TTI

The number of scheduling entities per transmission time interval (SE/TTI) describes the maximum number of UEs that the baseband is capable of scheduling in a single TTI.

The number of supported SE/TTI depends on the baseband model and the software version. For more information on the currently supported SE/TTI, see [Supported Capacity and Configurations for NR RAN](#).

2.3.2.5 Number of Supported CG/TTI

The number of configured grants per transmission time interval (CG/TTI) describes the maximum number of users using the configured grant feature that can be scheduled in a single TTI.

The maximum number of CG/TTI depends on the system constant `cgMaxResource`. The default value of this system constant is 50%, which means that up to 50% of the uplink processing capabilities that are allowed per slot and per cell (`maxNoOtaPuschSePerSlotPerCell`) can be used for configured grants. See Section 2.3.3.3.2 for more information.

The value of `cgMaxResource` can be tuned up to 80% for the specific case of Dedicate Networks.

It is important to note that the `cgMaxResource` system constant affects not only the uplink processing capabilities of the cell but also the maximum frequency resource (i.e., physical resource blocks) allocation for uplink configured grant users. See Section 2.3.3.3.3 for more information.

2.3.2.6 Carrier Bandwidth

The amount of required carrier bandwidth for time critical applications depends on the packet size, radio link quality and the number of users that will be simultaneously scheduled per TTI. Even for small packets, high bandwidth is required for cases with bad radio link quality or when a low coding rate is used to ensure that the message is correctly delivered without requiring retransmissions. See Section 2.3.3.4 for more information on bandwidth requirements.

2.3.2.7 Number of Cells and Distribution of Devices across Cells

The number of cells and the distribution of users can depend on:



- Available carrier bandwidth
- Latency requirements
 - Handover added delay
 - Maximum number of users with low latency requirements supported per cell
- Baseband redundancy requirements

In Section 2.4.2, general deployment and RAN features recommendations are given for the case that multiple cells are required.

2.3.2.7.1 Available Carrier Bandwidth

The number of cells is linked to the available carrier bandwidth, the required latency, and the supported number of users. If there is not enough bandwidth available to provide service to the UEs in defined area, the area and thus the load can be split in 2 and covered by 2 cells. See Section 2.3.3.4 for more information on bandwidth requirements.

The disadvantage of splitting a cell into multiple cells and reuse the bandwidth is the possibility of co-channel interference that can affect transmissions and produce retransmissions that will increase the latency.

2.3.2.7.2 Handover Added Delay

In the case that multiple cells are necessary, and a user is expected to move between cells, the delay (i.e., 50 ms for handover triggering time and 45 ms handover interruption time) introduced by the handover must be taken into the latency budget calculations. Moreover, features which are dependent on RRC configuration (e.g., configured UL grant) are expected to be temporarily unavailable during handover until the RRC Connection reconfiguration has been applied in the target cell. The time this takes varies between 10-16 ms.

2.3.2.7.3 Maximum Number of Low-latency Users Supported per Cell

Due to baseband capacity and control channel limitations a single cell is only capable of serving a maximum number of users with a given latency requirement. See Sections 2.3.3.1.7, 2.3.3.3.8 and 2.3.3.2.1 for more information on current capacities depending on the uplink scheduling method.

2.3.2.7.4 Baseband Redundancy Requirements

Baseband redundancy can be achieved by deploying multiple cells with overlapping coverage in separate basebands. In this case, the available bandwidth must be split among the overlapping cells to avoid co-channel interference issues. This type of setup also requires load awareness to equally distribute the traffic among the cells with overlapping coverage. Therefore, the NR Advanced Multi-Layer Coordination (MLC) feature is required so that the gNB can take into consideration cell load during traffic steering. Nevertheless, this feature does not consider latency requirements.



In case that the NR Advanced Multi-Layer Coordination (MLC) feature is used, it is recommended to disable periodic MLC to prevent handovers after the initial load distribution.

2.3.3 Dimensioning Methodology for Time Critical Low Bitrate Service

The two main requirements in time critical communications are bounded latency and reliability.

Reliability is achieved with retransmissions within the latency budget by targeting a specific block error rate (BLER).

The objective of this dimensioning exercise is to discover the maximum number of devices that the network can serve while maintaining the latency and reliability requirements of the time critical application and considering the network configuration and baseband processing limitations described in the previous section.

Moreover, because latency is such an important requirement features that affect transmission latency like uplink configured grant or prescheduling should also be considered.

Therefore, the maximum number of devices depends on:

- Time critical application requirements:
 - Bounded latency
 - Reliability
- 5G System parameters
 - TDD pattern
 - Number of SE/TTI or CG/TTI
 - Number of MAC (HARQ) retransmissions
 - Uplink scheduling configuration
 - Dynamic grant scheduling
 - Prescheduling
 - Uplink configured grant
 - Available carrier bandwidth
 - Link adaptation
- Traffic characteristics
 - Packet size
 - Traffic periodicity

In Sections 2.3.3.1, 2.3.3.2 and 2.3.3.2, three models that elaborate on the maximum user capacity are described considering the variables mentioned above. One model assumes dynamic grant scheduling, the other assumes prescheduling and the last model assumes that uplink configured grant is used.

These are simplified models which assume that the only traffic on the network belongs to TCC services. The evaluation of a system with mixed services (e.g., mobile broadband and time critical communications) is analyzed in Section 2.3.3.5.



2.3.3.1 Dimensioning for Dynamic Grant Scheduling

2.3.3.1.1 SR Periodicity

In dynamic grant scheduling before sending data in the uplink, a UE is required to send a SR.

Due to limitations in the capacity of the physical uplink control channel PUCCH, UEs are not allowed to send a SR at any moment. Instead, UEs have the opportunity of sending a SR at predefined intervals or SR periodicity.

After RRC establishment, UEs are assigned to a SR periodicity pool. In current products, there are three defined pools: shortest, medium, and long periodicity. The pool allocation is based on first come first served principle. First UEs to arrive are allocated to the shortest periodicity pool. Once this pool is full, new UEs are allocated to the medium periodicity pool. After the second pool is full, any new UE is allocated to the long periodicity pool.

RRC SR periodicity pools	Minimum number of users in the pool	SR periodicity
Lowest periodicity	5 users	minSrPeriod (depends on selected TDD pattern)
Medium periodicity	95 users	20 ms
Long periodicity		80 ms

Table 1 RRC SR periodicity pools per cell

In Table 1, the minimum number of users that can be allocated to each pool and the corresponding SR periodicity are shown.

The value for the minSrPeriod depends on the selected TDD pattern. The values for different TDD patterns are shown in Table 2.

TDD Patten	minSrPeriod in ms
DDDSU	5
DDDS1UUDS2UU	5

Table 2 Minimum SR period for different TDD patterns

PUCCH resources for SR transmission are allocated in terms of SR resources. The number of users that share a single SR resource depends on the SR periodicity and the selected TDD pattern. A summary of the UEs supported by a single SR resource is given in Table 3.

SR periodicity in ms	UEs per SR resource	
	DDDSU	DDDS1UUDS2UU
5	2	4



10	4	8
20	8	16
40	16	32
80	32	64

Table 3 Number of supported UEs per SR resource depending on SR periodicity and TDD pattern

According to Table 1, the minimum number of users in the lowest periodicity pool is 5 UEs and 95 UEs in the medium periodicity pool. The actual number of users in the lowest and medium periodicity pools will depend on the selected TDD pattern. As shown in Figure 6 and Table 3, for the DDDSU pattern a single SR resource can allocate 2 UEs. Therefore, 3 SR resources are required to allocate at least 5 UEs. As 3 SR resources can allocate 6 UEs in total, this will be the actual capacity of the lowest periodicity pool.

In a similar way, Figure 7 shows how for the DDDS1UUDS2UU a single SR resource can allocate 4 UEs and that 2 SR resources are required to allocate at least 5 UEs. Therefore, the lowest periodicity pool can support 8 UEs.

In Table 4, a summary of the number of users and SR periodicities for each pool and TDD pattern is given.

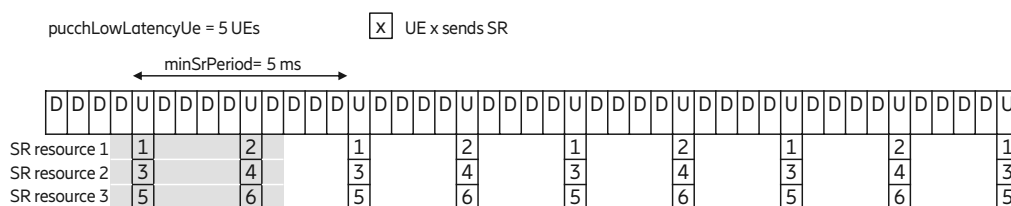


Figure 6 No. of users in lowest periodicity RRC SR periodicity pool for DDDSU pattern

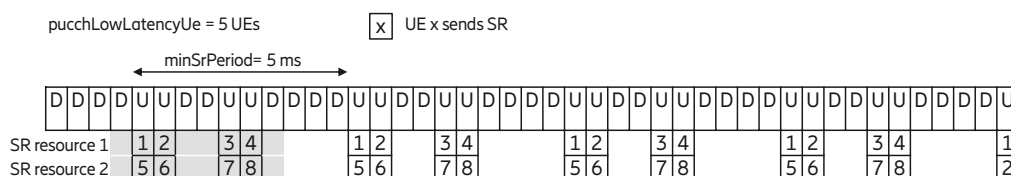


Figure 7 Number of users in lowest periodicity RRC SR periodicity pool for DDDS1UUDS2UU

In conclusion, depending on the time of arrival a UE can be assigned to the lowest, medium, or long SR periodicity pool. Users in the lowest SR periodicity pool may need to wait a maximum of 5 ms for an opportunity to send the SR. If the user is allocated to the medium SR periodicity pool, then the UE may need to wait up to 20 ms. Finally, users in the long periodicity pool may need to wait up to 80 ms to send a SR. The number of users in each pool depends on system constants and the selected TDD pattern.

TDD Pattern



		DDDSU	DDDS1UUDS2UU
Lowest RRC SR periodicity pool	SR Periodicity	5	5
	No. of users	6	8
Medium RRC SR periodicity pool	SR Periodicity	20	20
	No. of users	96	96
Long RRC SR periodicity pool	SR Periodicity	80	80

Table 4 Summary of SR periodicity and number of users in each RRC SR periodicity pool

2.3.3.1.1.1 NR Service-Adaptive SR Periodicity (FAJ 121 5595)

As described in Section 2.3.3.1.1, the SR periodicity is assigned to a UE on a first come first served basis. The NR Service-Adaptive SR Periodicity feature allows to configure different SR periodicities per UE group ([UE Grouping Framework](#)) and to configure additional SR periodicity pools. Therefore, it is possible to reserve the pools with short SR periodicity for UEs in a specific UE group.

The feature removes the limitation of having only three pools (lowest, medium, and long SR periodicity). Instead, now up to 6 SR periodicity pools are available. For the first 5 pools, the SR periodicity, the number of SR resources and thus the maximum number of users per pool can be set. The 6th pool has the longest SR periodicity, and it is not configurable.

A total of 56 SR resources can be distributed among the 5 configurable SR periodicity pools, while 36 SR resources are allocated to the 6th pool which is not configurable. In case that the initial 56 SR resources are not completely allocated to the first 5 configurable pools, the remaining resources are allocated to the 6th default pool. This assumes that the cell does not support high speed UEs.

The allocation of SR resources in the pools affects the maximum number of connected UEs in the cell. If more SR resources are given to pools with a short SR periodicity value than to pools with long SR periodicity, the maximum number of RRC Connected UEs for such a cell can decrease below the maximum number of RRC Connected UEs per cell defined for the baseband in the [Supported Capacity and Configurations for NR RAN](#).

In Table 3, the maximum number of UEs that can be supported in the lowest periodicity pool using a DDDSU TDD pattern with 5 ms SR periodicity is 6. However, as shown in the example of Table 5, using the NR Service-Adaptive SR Periodicity feature, it is possible to define a RRC SR periodicity pool with SR periodicity 5 ms (Pool 1) and the maximum number of UEs will depend on the configured number of SR resources. In the example, by allocating 20 SR resources to this pool, up to 40 UEs can be supported with 5 ms SR periodicity.

Number of users



	DDDSU	DDDS1UUDS2UU
RRC SR Periodicity Pool 1 SR Periodicity=5 ms SR resources=20	20x2=40	20x4=80
RRC SR Periodicity Pool 2 SR Periodicity=20 ms SR resources=20	20x8=160	20x16=320
RRC SR Periodicity Pool 3 SR Periodicity=40 ms SR resources=16	16x16=256	16x32=512

Table 5 Example of RRC SR Periodicity pools using the NR Service-Adaptive SR Periodicity feature

The NR Service-Adaptive SR Periodicity feature allows to increase the number of supported UEs with a low SR periodicity at the expense of supporting less UEs in total per cell. Moreover, periodicity pools can be reserved to specific UE groups.

For more information, see [UE Grouping Framework](#).

2.3.3.1.1.2 QoS Controlled SR Scheduling

QoS Controlled SR Scheduling is a function of the [QoS Framework](#) (FAJ 121 5157) that can be used to change the configuration of the SR uplink scheduling priority based on the QoS of the UE.

Without this feature, SR scheduling has an uplink scheduling weight higher than any data radio bearer (DRB) and thus will likely be scheduled first.

After activating the feature, the SR initial uplink weight will be equal to the highest priority DRB of the UE and increase over time.

In high load scenarios, this feature helps to maintain the QoS of high priority DRBs like voice bearers and prevents SR to hog all resources.

In the specific case of a factory environment, this feature is useful if packet segmentation cannot be avoided, and it is expected. If packet segmentation occurs, the UE sends buffer status reports (BSR) to the gNB to request additional resources. However, the BSRs can be down prioritized due to the large number of pending SRs. If the BSRs are not scheduled, due to SRs having a higher priority, the UE cannot get the necessary resources and the packets are delayed. Therefore, by activating this feature the priority of the SRs can be lowered so that SRs and BSRs are treated in a similar way.

The feature can be applied to a specific set of UEs through the UE Grouping Framework.



For more information, see [QoS Framework](#).

2.3.3.1.2 Control Channel Limitations using Dynamic Grant

The physical downlink control channel (PDCCH) in NR carries Downlink Control Information (DCI). DCI includes resource scheduling information for the PDSCH and PUSCH, uplink power control indication, slot format indication and more.

The UE requires the PDCCH information to locate in time and frequency the information transmitted in the PDSCH and to assemble and send the PUSCH data. Therefore, reliable decoding of PDCCH is very important for the overall performance of the system.

The basic resource of the PDCCH is the control channel element (CCE). Assuming a 100 MHz carrier bandwidth and a single OFDM symbol for PDCCH, 45 CCEs are available for transmission (See the [Control Channel Guideline NR](#) for information on the PDCCH capacity for other carrier bandwidths). Inside of these CCEs, the UE search space (i.e., UE specific and common search space) that contains the control information (e.g., DCI message) that the UE requires is placed.

However, depending on the aggregation level used, the actual number of CCEs required for PDCCH transmission increases. There are 5 aggregation levels available. In aggregation level 1, 1 CCE is required to carry 1 PDCCH message- (e.g., DCI message) and in each subsequent level the number of required CCEs increases:

- Aggregation level 1: 1 CCE
- Aggregation level 2: 2 CCEs
- Aggregation level 4: 4 CCEs
- Aggregation level 8: 8 CCEs
- Aggregation level 16: 16 CCEs

The selection of the aggregation level depends on the radio channel SINR measurements. In bad channel conditions (i.e., low SINR), a high aggregation level with more CCEs is used to allow for more robust encoding of the PDCCH information to ensure that the message is correctly decoded. However, the disadvantage is that there are less CCEs available for additional PDCCH transmissions. This can cause a delay in the transmission of control messages increasing the latency.

In the case of a high SINR (i.e., good radio channel conditions), a lower aggregation level is used and thus less CCEs resources are required per PDCCH message. As there are more available resources, more simultaneous PDCCH messages are possible than in the low SINR case.

In summary, good SINR levels are necessary to ensure that a low aggregation level and that CCE resources are efficiently used. If the SINR levels are low, the CCEs are used to increase the robustness of the PDCCH at the cost of decreasing its capacity. This can cause increase latency as UEs must wait for longer to receive the necessary control information.



2.3.3.1.2.1 NR Service-Adaptive Link Adaptation (FAJ 121 5513)

The feature [NR Service-Adaptive Link Adaptation](#) adds the new MoM parameter `pdccchLaSinrUeOffset` which allows to set a fixed offset in the PDCCH SINR calculation.

The PDCCH SINR is calculated based on channel quality indicator (CQI) reports from the UE and a constant SINR offset (`pdccchLaSinrUeOffset`). Based on the PDCCH SINR, the appropriate aggregation level for the PDCCH is selected.

By modifying this SINR offset, it is possible to increase the channel robustness. As an example, selecting a negative offset will decrease the estimated PDCCH SINR and thus a high aggregation level will be selected. As explained in the previous section, high aggregation levels consume more resources (i.e., CCEs) but are more resilient to radio channel problems.

2.3.3.1.3 SR-to-transmit Delay

After the UE sends the SR and receives the uplink grant from the gNB, it must wait until uplink resources are available. The interval between the SR and the transmit time is the SR-to-transmit delay.

This delay depends on the number of UEs, the selected TDD pattern and the SE/TTI capacity of the baseband.

Based on the current product capabilities (e.g., processing delays), in this guideline a minimum SR to transmit of 5 ms is assumed.

The SR to transmit delay can be longer in the case of aperiodic CSI transmissions but this is out of the scope of this guideline.

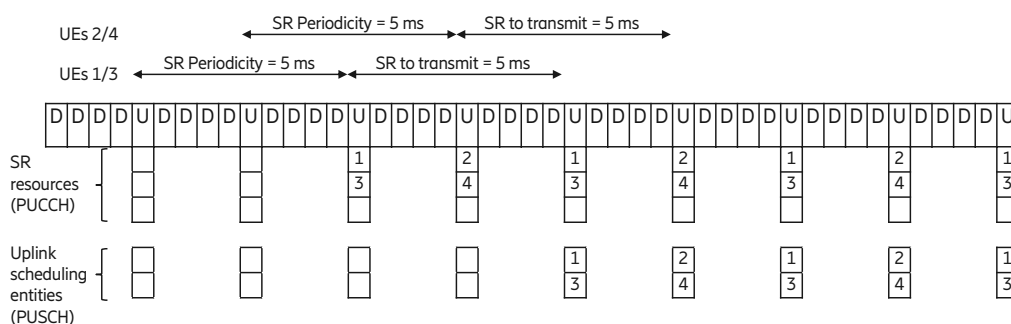


Figure 8 SR-to-transmit delay

In Figure 8, a simplified example of uplink scheduling is shown. In this example, it is assumed that there are 4 UEs in the system. All UEs are in the RRC SR periodicity pool of 5 ms. Moreover, the baseband has a 2 SE/TTI capacity and a DDSU pattern was selected. After sending the SR, the fastest that the UE can transmit is 5 ms.

However, if an additional UE is added, as shown in Figure 9, the SR-to-transmit delay will increase as the system does not have enough capacity (i.e., SE/TTI) to



transmit 3 UEs in a single uplink slot. Therefore, the SR-to-transmit delay for some of the UEs is now increased to 7.5 ms.

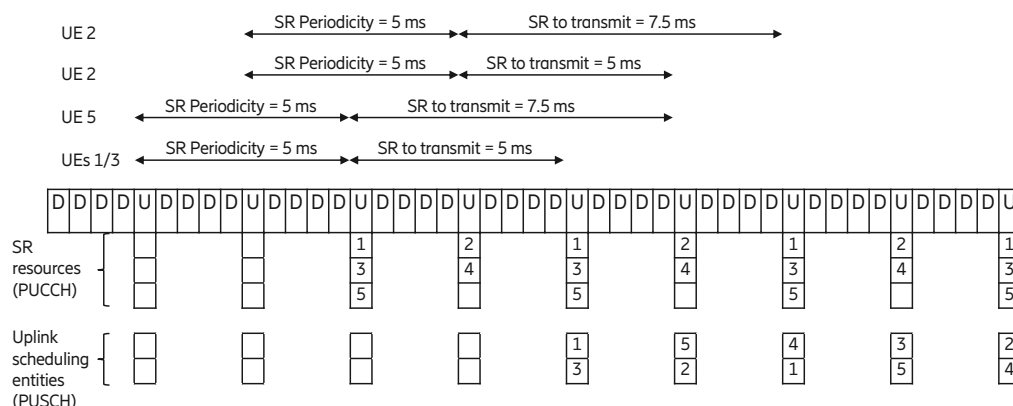


Figure 9 Number of users effect on SR-to-transmit delay

For the analytical model that follows, the minimum SR-to-transmit delay of 5 ms is assumed. As shown in Figure 8, this assumption is valid if there is no congestion in the PDCCH and the PUSCH.

2.3.3.1.4 RAN L1 Processing Delays

RAN layer 1 processing delays include delays in both the UE and the gNB. For example, after the reception of the SR, the gNB requires time to encode and send the uplink grant. In the same way, the UE requires time to decode and process the PUSCH.

The processing delays depend on the traffic load and the capacity of the products. This capacity can improve in the future. Nevertheless, for the purpose of this guideline it will be assumed that the sum of UE and gNB processing delays equals to 1.5 ms.

This estimation assumes:

- Only layer 1 delays
- UE processing delay of 0.5 ms
- gNB processing delay
 - 1 ms for PUSCH decoding
 - 1.5 ms for uplink grant encoding

2.3.3.1.5 MAC (HARQ) Retransmissions

In case that a packet sent by the UE cannot be correctly decoded by the gNB, a retransmission will be triggered as part of the Hybrid automatic repeat request (HARQ) process.

Each retransmission comes at the expense of additional scheduling resources and increased delay.

Depending on the TCC service latency requirements, a certain number of retransmissions are acceptable. Therefore, these retransmissions must be included in the delay and capacity calculations.

In this guideline, it will be assumed that a single retransmission adds 5 ms to the total delay. An example of how this value is estimated is shown in Figure 10.

After the UE transmits the data in the PUSCH channel, the gNB requires 1 ms to decode and verify the integrity of the received packet. After detecting failure, the gNB needs to encode and send an uplink grant indicating the UE that it must retransmit. Uplink grants are sent two slots before the PUSCH ($k_2=2$) and the gNB requires 1.5 ms to encode the grant. Therefore, the retransmission cannot occur in the immediate uplink slot after the failure was detected and occurs one uplink slot later.

The retransmission example shown in Figure 10 is also valid for retransmissions in prescheduling or uplink configured grant.

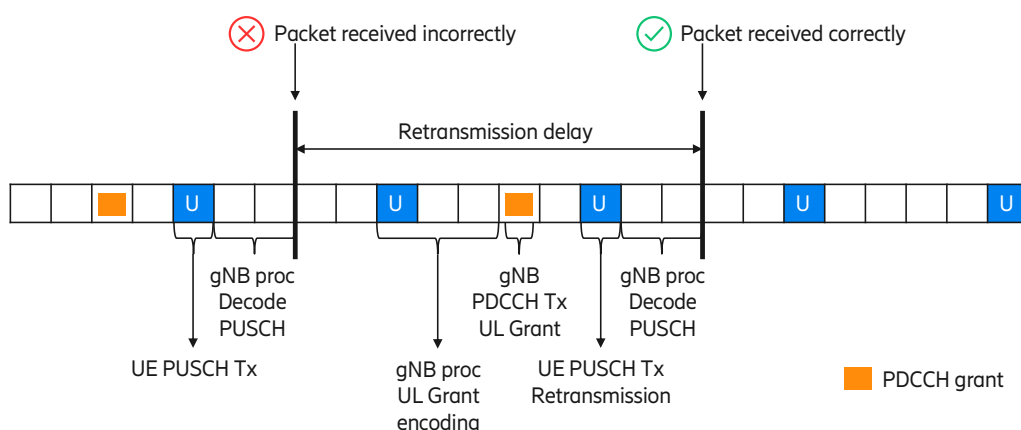


Figure 10 Example of retransmission delay calculation

It is recommended to reserve part of the cell SE/TTI capacity to account for possible retransmissions. A rule of thumb is to reserve 10% of the total SE/TTI capacity of the baseband for retransmissions assuming a 10% block error rate (BLER).

2.3.3.1.5.1 NR Service-Adaptive Link Adaptation (FAJ 121 5513)

This feature allows to customize link adaptation for different UE groups in both uplink and downlink.

Using this feature, it is possible to have a custom link adaptation and specify the following parameters for uplink adaptation:

- BLER target
- Maximum modulation and coding scheme (MCS) index,
- Minimum transport block size (TBS)
- HARQ mode



- SR grant size
- ReTx BSR timer

In the downlink the following parameters can be adjusted:

- BLER target
- Maximum modulation and coding scheme (MCS) index,
- Maximum transport block size (TBS)
- HARQ mode
- PDCCH SINR offset

These parameters can be applied to a specific UE group defined using the [UE Grouping Framework](#).

This feature is important to adjust the block error rate (BLER) target and the SR grant size.

Lowering the BLER target (e.g., changing from a 10% to 0.1% BLER) improves the probability of a successful transmission and reduces the probability of retransmissions. To achieve this, the network will choose a more conservative, namely lower, MCS index to improve the odds of successful decoding at the receiver. Thus, less retransmissions will be required and therefore, the mean delay will be reduced. The increased reliability will come at the expense of a reduced system capacity as more resources will be dedicated to a single user because of the low MCS index.

The SR grant size must also be adjusted so that it matches the transmitted packet size and therefore, avoid segmentation. If this parameter is not adjusted, the packet will be segmented, and the UE will be required to request additional resources to send the remaining fragments. This will increase the latency and add more load the PDCCH and PDSCH.

It is important to note that if the SR grant size is larger than the actual packet size, the UE will add extra padding to match the SR grant size. This increases the utilization of PUSCH resources.

For more information, see [NR Service-Adaptive Link Adaptation](#)

2.3.3.1.6 Analytical Model

The following model will provide the maximum number of users that can be served using dynamic grants for a given latency.

Definition of key parameters:

- Latency target, l [ms]
- Traffic periodicity, p_T [ms]
- t_{BS} = gNB processing delay
- t_{UE} = UE processing delay
- t_{slot} = slot duration



- n = number of retransmissions
- t_{retx} = HARQ retransmission delay
- $t_{SR-to-tx}$ = SR-to-transmit delay (excluding queuing)
- l' = budget for SR alignment and queuing delay
- U is the number of UL slots within l'
- s is the number of users scheduled per slot (or per TTI)

The budget for SR alignment and queuing delay can be calculated as follows:

$$l' = l - t_{SR-to-tx} - t_{BS} - t_{UE} - t_{slot} - n \cdot t_{retx} [\text{ms}]$$

The number of users scheduled per slot is calculated as follows:

$$s = SE/TTI \text{ limitation} - SE/TTI \text{ used for retransmission}$$

SE/TTI limitation is equal to the supported SE/TTI that can be found in [Supported Capacity and Configurations for NR RAN](#).

The number of slots within l' depends on the selected TDD pattern

- $U = \frac{l'}{2.5 \text{ ms}}$ (DDDSU) 1 uplink slot every 2.5 ms
- $U = \frac{l'}{5 \text{ ms}} \cdot 4$ (DDDS1UUDS2UU) 4 uplink slots every 5 ms

Finally, the maximum number of devices C supported for a specific latency target l is calculated as

$$C = U \times s$$

2.3.3.1.7 Results

Based on the analytical model described before and on the following assumptions:

- RRC SR periodicity pools as defined in Table 1 (lowest, medium, and longest SR periodicity pools),
- maximum number of RRC connected users per cell = 1500,
- $t_{BS} + t_{UE} + t_{slot} = 2 \text{ ms}$,
- 0 application packets can arrive after the latency budget, and
- every packet is transmitted in a single uplink slot

the maximum number of devices according to the requested RAN latency for different SE/TTI and TDD patterns is calculated in Figure 11, Figure 12, Figure 13 and Figure 14.

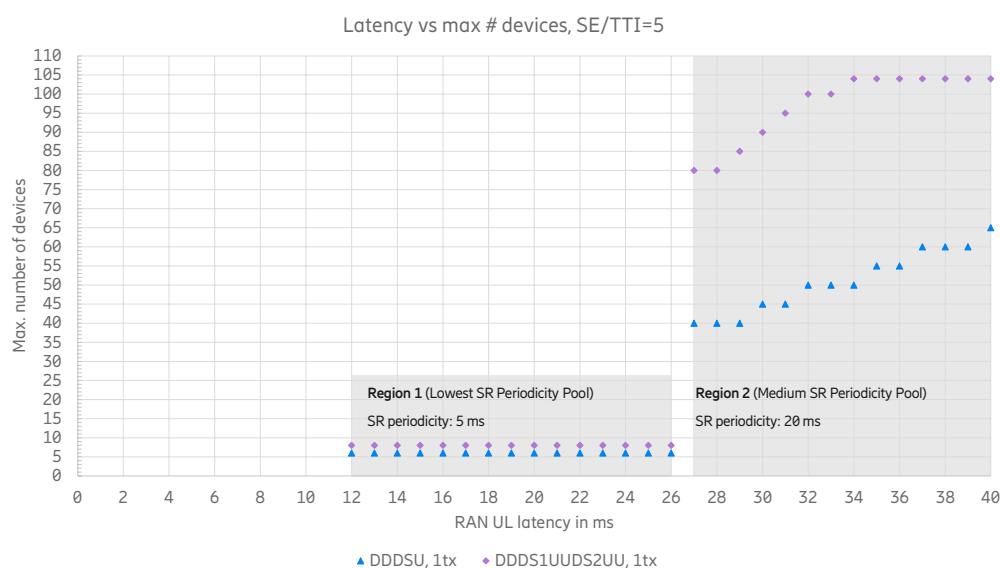


Figure 11 Maximum number of devices per cell for a SE/TTI=5

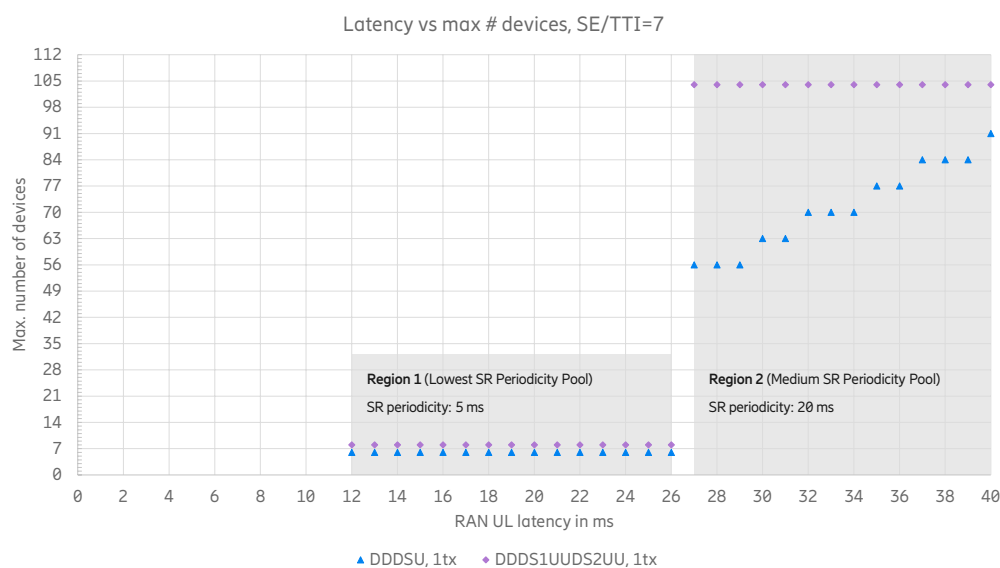


Figure 12 Maximum number of devices per cell for a SE/TTI=7

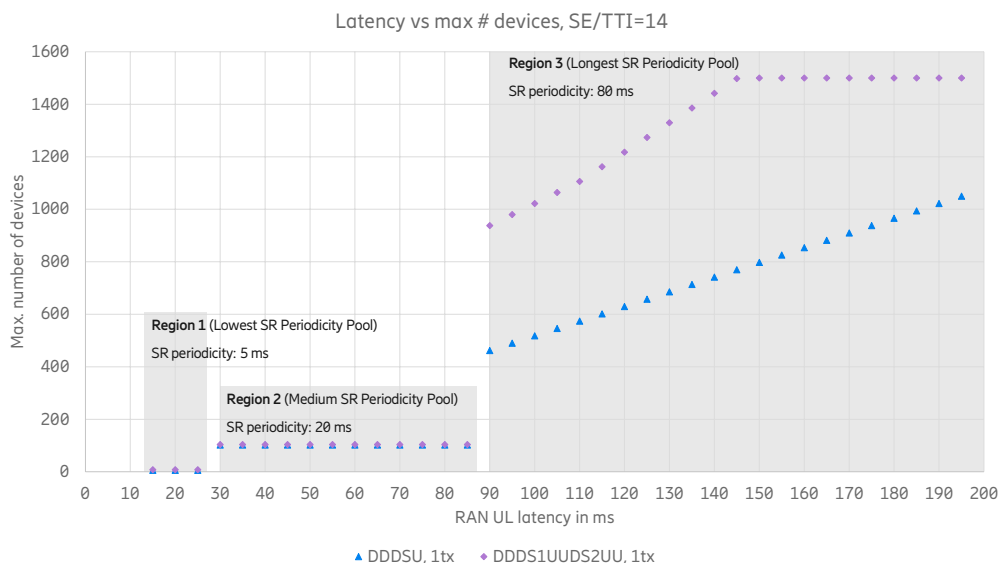


Figure 13 Maximum number of devices per cell for a SE/TTI=14

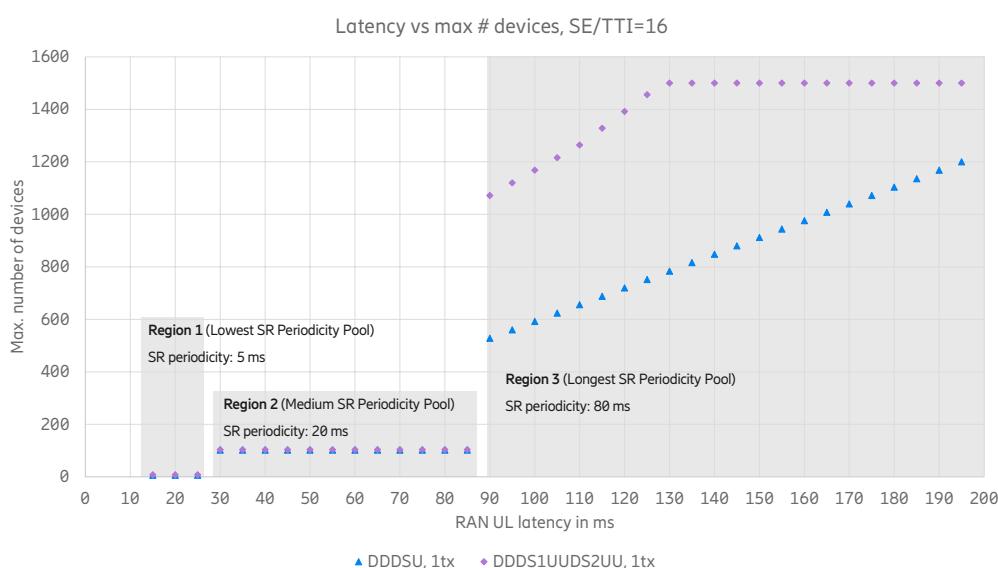


Figure 14 Maximum number of devices per cell for a SE/TTI=16

From these results, the following conclusions can be made:

- The minimum RAN latency supported by both TDD patterns under these conditions is 12 ms. This includes 5 ms delay due SR periodicity, 5 ms delay for SR to transmit (minimum delay possible) and 2 ms of processing delays in gNB and the UE.
 - If support for a lower latency is required, a TDD pattern with a shorter min SR period (e.g., DDSU) could be selected. Otherwise, another scheduling scheme such as prescheduling or uplink configured grant can be used instead of dynamic scheduling.



- The graphs can be divided in 3 regions according to the active RRC SR periodicity pools:
 - A latency of at least 12 ms is required to allocate users in the pool with the lowest SR periodicity (5 ms SR periodicity)
 - A latency of at least 27 ms is required to allocate users in the pool with the medium SR periodicity (20 ms SR periodicity)
 - A latency of at least 87 ms is required to allocate users in the pool with the longest SR periodicity (80 ms SR periodicity)
- The maximum capacity that can be achieved in each of these 3 regions depends on the maximum number of users that can be allocated in each of the 3 pools:
 - Region 1: 6 UEs for DDDSU and 8 UEs for DDDS1UUDS2UU
 - Region 2: $6+96=102$ UEs for DDDSU and $8+96=104$ UEs for DDDS1UUDS2UU
 - Region 3: 1500 UEs which is the maximum number of RRC connections per cell ([Supported Capacity and Configurations for NR RAN](#))
- In region 1, capacity is limited by the maximum number of UEs that can be allocated in the lowest periodicity pool.
 - Using the NR Service-Adaptive SR Periodicity feature (described in Section 2.3.3.1.1.1), the number of users that can access this pool can be increased to have more capacity for low latency users.
- In regions 2 and 3, the capacity is not only limited by the maximum number of users in the medium and longest RRC SR periodicity pools but also by the SE/TTI and the uplink capacity of the TDD pattern itself.
 - As the latency requirements are relaxed, more slots are available for transmission and thus the capacity increases.
 - E.g., assuming a latency of 27 ms, 8 uplink slots are available for transmission in the DDDSU pattern. If a longer latency of 30 ms can be supported, then an additional uplink slot is available for a total of 9 uplink slots.
 - The SE/TTI determines how many UEs can be scheduled in a single slot. Therefore, there is a linear relation between capacity and SE/TTI.
- The SE/TTI graph should be selected based on the baseband capacity ([Supported Capacity and Configurations for NR RAN](#)).

Depending on the BLER target and the channel conditions retransmissions can be expected. In Figure 15, the effect of multiple retransmissions in the capacity curves can be seen. Assuming a 5 ms added delay for retransmissions (Section 2.3.3.1.5), the original graph from Figure 12 (without considering retransmissions) is shifted to the right by n times 5ms, where n is the number of retransmissions.

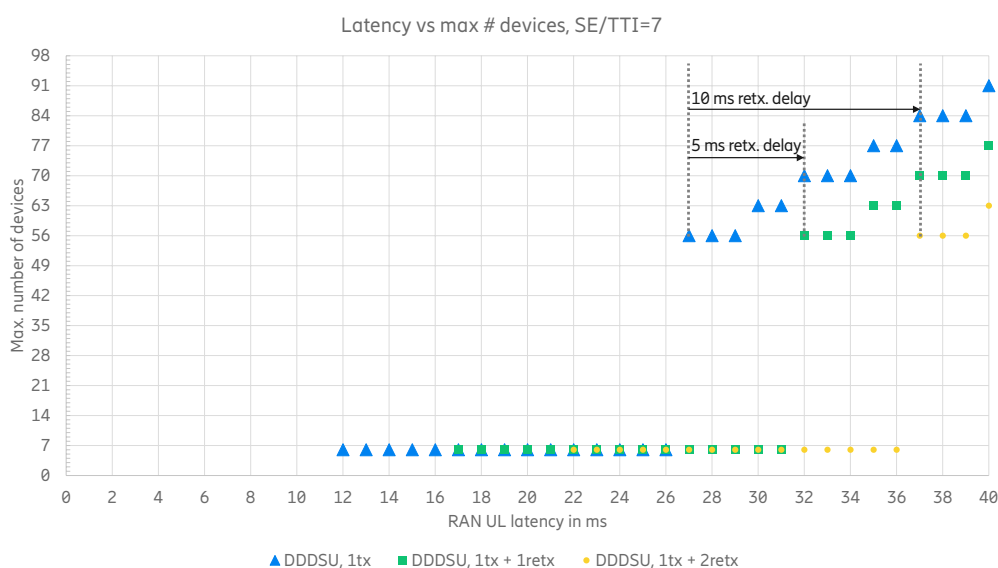


Figure 15 Maximum number of devices per cell for a SE/TTI=7 with 1 and 2 retransmissions

The NR Service-Adaptive SR Periodicity feature (described in Section 2.3.3.1.1.1) can be used to change the size of the default RRC SR periodicity pools.

In Figure 16, it can be seen the effect in the capacity of region 1 by using a single customized RRC SR periodicity pool as shown in Table 6.

	Number of users	
	DDDSU	DDDS1UUDS2UU
RRC SR Periodicity Pool 1	56x2=112	56x4=224
SR Periodicity=5 ms		
SR resources=56		

Table 6 Example of customized RRC SR periodicity pool with the NR Service-Adaptive SR Periodicity feature

Compared to Figure 11, more users are supported in the region 1 as the capacity of this region is no longer limited by the maximum number of users in the lowest RRC SR periodicity pool.

Comparing Figure 16 and Figure 17 shows that the capacity is now limited by the SE/TTI. By increasing the SE/TTI from 5 to 15, the maximum capacity of the first pool is achieved for both TDD patterns.

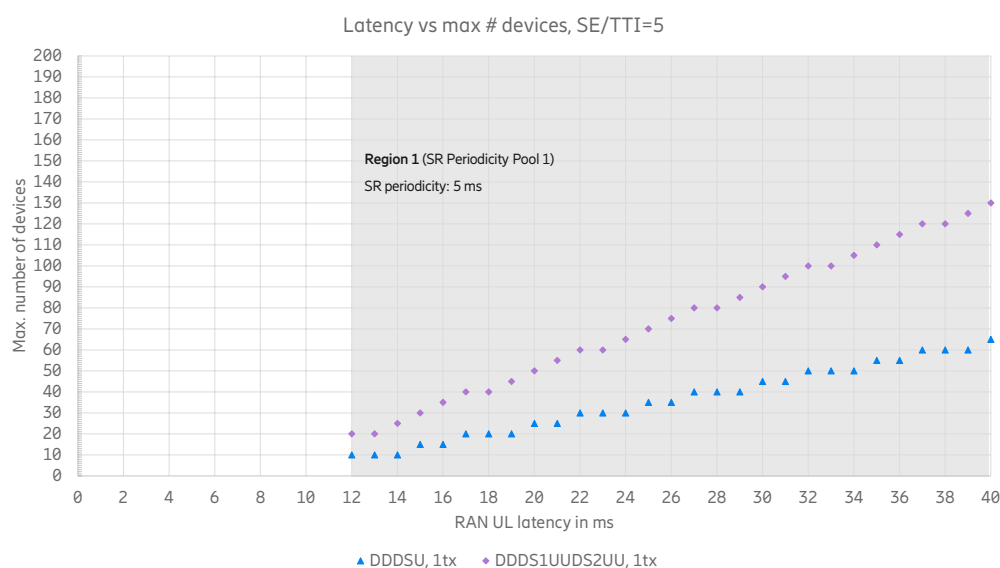


Figure 16 Maximum number of devices per cell for a SE/TTI=5 and using NR Service-Adaptive SR periodicity

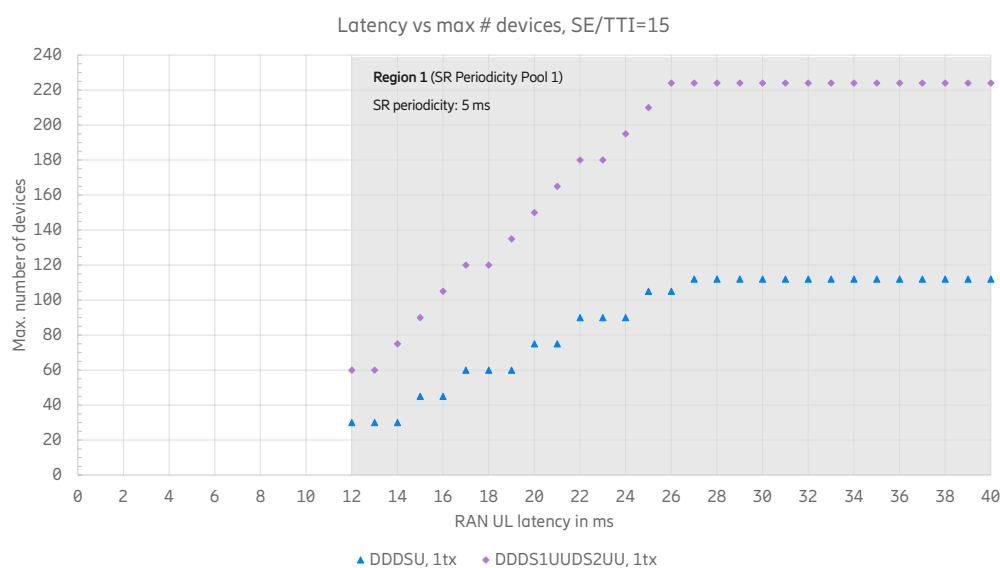


Figure 17 Maximum number of devices per cell for a SE/TTI=15 and using NR Service-Adaptive SR periodicity

2.3.3.2 Dimensioning for NR Service-Adaptive Prescheduling

The feature NR Service-Adaptive Prescheduling has two modes:

- Basic priority
- High priority



UEs configured with basic priority prescheduling get lower priority than dynamic grant UEs or than high priority UEs.

UEs configured with high priority get higher priority than dynamic grant UEs and basic priority UEs. However, there is a capacity limitation of a single high-priority grant per uplink slot and 2 high priority UEs per capacity module.

Due to the capacity limitation for high-priority prescheduling, it is preferred to use basic-priority prescheduling.

Uplink prescheduling periodicity is equal to the uplink slot periodicity of the TDD pattern. For example, for the DDDSU TDD pattern, the periodicity is 2.5 ms.

In the same way as dynamic grant scheduling, the number of users that can be scheduled in a single uplink slot depends on the SE/TTI capacity of the baseband ([Supported Capacity and Configurations for NR RAN](#)).

Uplink latency for a single prescheduled UE is approximately 4.5 ms in the case of a single transmission and 9.5 ms in the case that 1 retransmission is required.

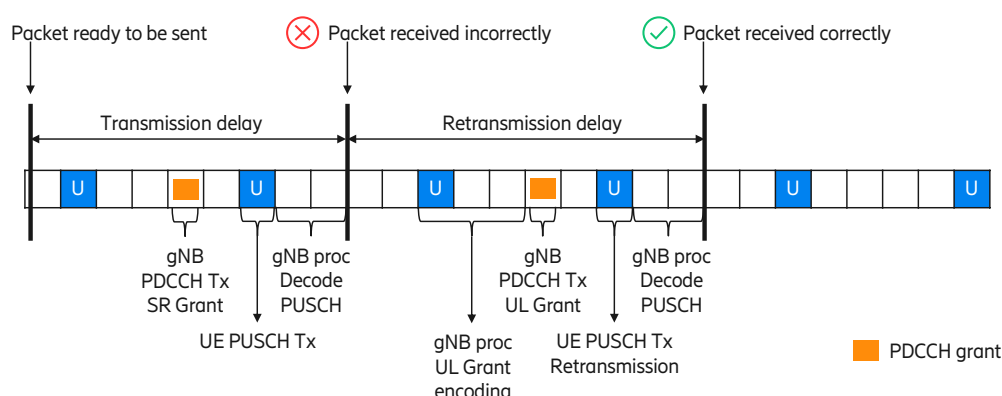


Figure 18 Transmission delay for NR prescheduling

Figure 18 shows how the transmission delay of 4.5 ms is calculated for prescheduling. This is a worst-case scenario in which the UE is ready to transmit just before the uplink slot and therefore there is not sufficient time to prepare the packet and the UE needs to wait for the next uplink slot in ~ 3 ms. After that, the UE transmits and then the gNB takes 1 ms to decode and process the packet.

To avoid segmentation, the packet should fit in the resources allocated to the UE in the uplink grant sent by the gNB. Therefore, the `preschedulingDataSize` MoM should be modified with the expected packet size also including headers.

2.3.3.2.1 Results

Based on the latencies described before and on the following assumptions:

- only basic priority users,
- single service (e.g., no MBB users using dynamic grant uplink scheduling),



- $t_{BS} + t_{UE} + t_{slot} = 2 \text{ ms}$ (where t_{BS} = gNB processing delay, t_{UE} = UE processing delay and $t_{slot} = 0,5 \text{ ms}$),
- retransmission delay 5 ms,
- 0 application packets can arrive after the latency budget, and
- every packet is transmitted in a single uplink slot

the maximum number of devices according to the requested latency for SE/TTI=7 and TDD pattern DDDSU is calculated in Figure 19.

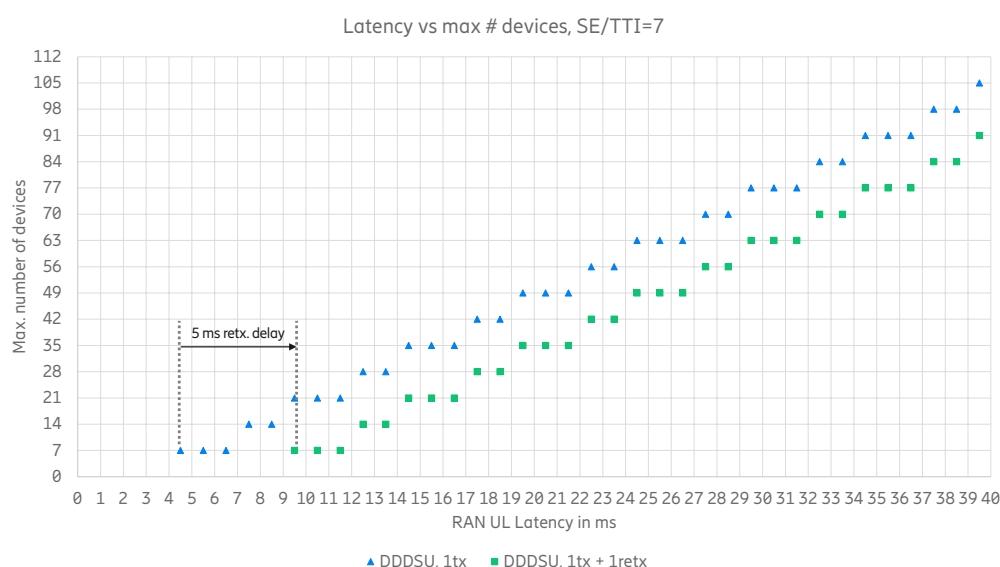


Figure 19 Maximum number of devices per cell for SE/TTI=7 using NR Service-Adaptive Prescheduling

From these results, the following conclusions can be made:

- The minimum RAN latency supported under these conditions is 4.5 ms. This includes 3 ms to wait for an uplink slot, 0.5 ms for slot alignment during the transmission and 1 ms of processing delays in gNB.
- Compared to dynamic grant scheduling lower latencies can be achieved (e.g., 12 ms is the lowest latency achievable with dynamic grant)
- Capacity is dependent on the SE/TTI capacity of the baseband

2.3.3.3 Dimensioning for Uplink Configured Grant for URLLC Dedicated RAN Scheduling

2.3.3.3.1 Uplink Configured Grant Periodicity

In configured grant scheduling, the main factor that affects the latency is the selected periodicity. This parameter determines how often can UE transmit in the uplink. Available periodicities are 2.5 ms, 5 ms, 10 ms, 20 ms, 40 ms and 80 ms among others (See [Uplink Configured Grant for URLLC Dedicated RAN](#) for a complete list of the supported periodicities). The selection of the periodicity depends on the selected TDD pattern. Figure 20 and Figure 21 show some of the supported periodicities in the DDDSU and DDDS1UUDS2UU TDD patterns.



Only a single configured grant periodicity is supported in a gNB.

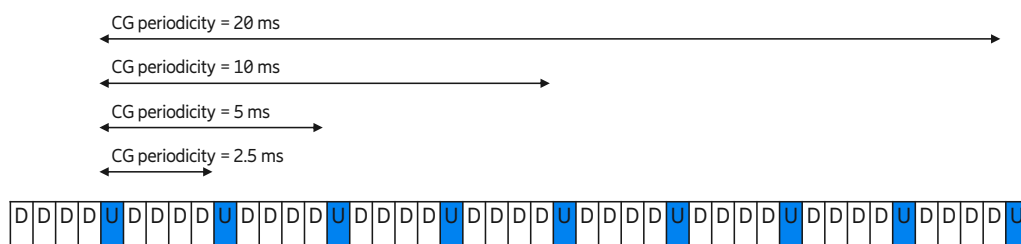


Figure 20 Configured grant periodicities for DDDSU TDD pattern

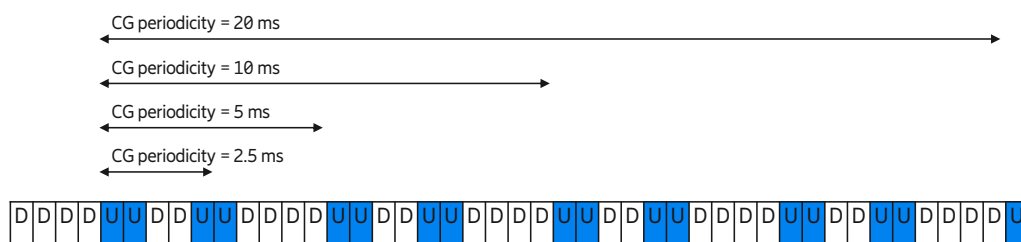


Figure 21 Configured grant periodicities for DDDS1UUDS2UU TDD pattern

2.3.3.3.2 Uplink Configured Grant Capacity (CG/per TTI)

During the uplink scheduling process there are two system constants involved that must be understood. The `maxNoPuschSePerSlotAllocator` which is referenced in the [Supported Capacity and Configurations for NR RAN](#) describes the maximum scheduling capacity per cell. The second system constant is called `maxNoOtaPuschSePerSlotPerCell` and refers to the uplink processing capability of the cell.

In the case of users using dynamic grant scheduling, both system constant limits apply as the UE must be first scheduled and then processed once the data is received.

However, in the case of users using the uplink configured grant feature, the first system constant (`maxNoPuschSePerSlotAllocator`) does not apply as these users do not require scheduling capacity. Instead, uplink configured grant users are only limited by the uplink processing capability of the cell (`maxNoOtaPuschSePerSlotPerCell`).

The configured grant capacity, i.e., number of configured grants per TTI, is controlled by the system constant `cgMaxResource`. The default value of this system constant is 50%, which means that up to 50% of the uplink processing capability of the cell (`maxNoOtaPuschSePerSlotPerCell`) can be used for configured grants. This system constant has a range of 0-80%, thus at most, 80% of uplink SE/TTI per cell resources can be allocated for configured grants.



There is also a limit in the configured grant capacity per baseband capacity module. This is calculated using the same system constant `cgMaxResource` and multiplying it by the system constant `maxNoOfOtaPuschSePerSlotPerBbm` which represents the maximum uplink processing capability of the baseband capacity module.

cgMaxResource	OTA SE/TTI	CG/TTI	maxOtaPerTtiPerBbm	maxNrofCgPerTtiPerBbm
50%	16	$16 \cdot 50\% = 8$	42	$42 \cdot 50\% = 21$
80%	16	$16 \cdot 80\% = 12$	42	$42 \cdot 80\% = 33$

Table 7 Configure grant per TTI capacity per cell and per baseband (C1-XBM2-TDD)

Table 7 shows that by using the default value of 50% for `cgMaxResource` a maximum of 8 CG/TTI (assuming that `maxNoOtaPuschSePerSlotPerCell` is equal to 16) can be supported per cell and up to 21 per baseband capacity module (assuming that `maxOtaPerTtiPerBbm` is equal to 42).

In the specific case of Dedicated Networks, the system constant `cgMaxResource` can be increased up to 80%. In this case, up to 12 CG/TTI can be supported by cell and 33 CG/TTI can be supported by baseband capacity module. This assumes that `maxNoOtaPuschSePerSlotPerCell` is equal to 16 and that `maxOtaPerTtiPerBbm` is equal to 42.

It is important to note that the system constant `cgMaxResource` affects both the uplink processing capabilities and the maximum frequency resources (i.e., physical resource blocks) allocation (See Section 2.3.3.3.3 for more information).

2.3.3.3.3 Uplink Configured Grant Bandwidth

The maximum number for physical resource blocks (PRBs) that will be allocated to uplink configured grant users is controlled by the system constant `cgMaxResource`.

As mentioned in Section 2.3.3.3.2, the default value of this system constant is 50%. Therefore, up to 50% of the available PRBs can be allocated to UEs using uplink configured grant.

In the case of dedicated networks, the system constant can be increased up to 80%. Thus, at the most, 80% of PRBs can be allocated to UEs using uplink configured grant.

It is important to note that the system constant `cgMaxResource` affects both the maximum frequency resources (i.e., physical resource blocks) allocation and the uplink processing capabilities and (See Section 2.3.3.3.2 for more information).



2.3.3.3.4 Uplink Configured Grant Data Size

The configured grant data size is a MoM parameter that defines how much data can be transmitted in one configured grant occasion.

The data size estimate should also include a margin for headers to transmit the complete packet without segmentation. The size of the headers is dependent on the configuration. As an example, 68 bytes should be considered for IPv6 without ROHC to consider the TCP/IP, PDCP and RLC headers.

The maximum size of packet sent using configured grant without segmentation is 10000 bytes (See [Uplink Configured Grant for URLLC Dedicated RAN](#)).

2.3.3.3.5 Uplink Configured Grant Modulation and Coding Scheme

Link adaptation is not available in the Uplink Configured Grant for URLLC Dedicated RAN feature and the MCS configuration for configured grant is defined by the system constant cgMCS. The default value for this system constant is MCS0 to provide high reliability for most users.

The MCS should be selected according to the SINR conditions.

2.3.3.3.6 Capacity Calculation

The maximum number of supported users depends on:

- CG periodicity
- CG per TTI
- TDD pattern

As shown in Figure 22, a limited number of users can be supported for CG periodicity of 2.5 ms using the DDDSU TDD pattern. In this periodicity, the maximum number of supported users is dependent on the CG/TTI capacity. As an example, with CG/TTI=3 a total of 3 UEs can be supported but with CG/TTI=8 a total of 8 UEs can be supported.

If a UE requests uplink configured grant activation after the capacity limit is achieved, it will be scheduled using dynamic grant instead. This must be considered while planning the maximum number of UEs that belong to a UE group ([UE Grouping Framework](#)) that will use the Uplink Configured Grant for URLLC for Dedicated Networks feature. If the number of UEs in the group is larger than the configured grant capacity limit, some of the UEs in the group will end up being scheduled using dynamic grant increasing their RAN latency.

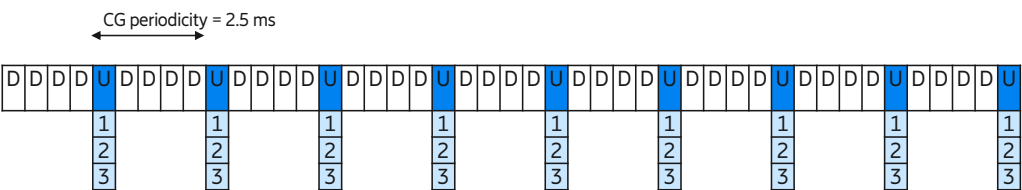




Figure 22 Uplink Configured Grant for URLLC Dedicated RAN capacity for CG periodicity=2.5 ms, CG/TTI=3 and TDD pattern DDDSU

In Figure 23, the CG periodicity is increased 4 times, compared to Figure 22, to 10 ms. By increasing the CG periodicity, the number of supported users is also increased 4 times.

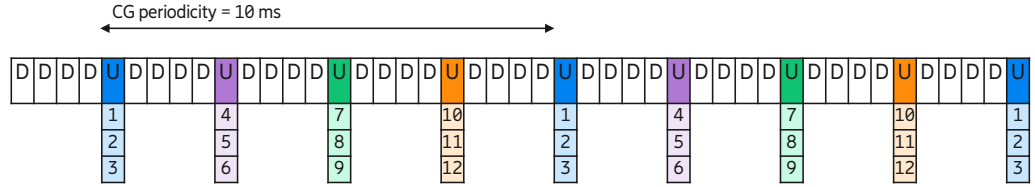


Figure 23 Uplink Configured Grant for URLLC Dedicated RAN capacity for CG periodicity=10 ms, CG/TTI=3 and TDD pattern DDDSU

The effect of the TDD pattern selection is shown in Figure 24. Under the same conditions used for Figure 23, the capacity is doubled by using a TDD pattern with double uplink capacity.

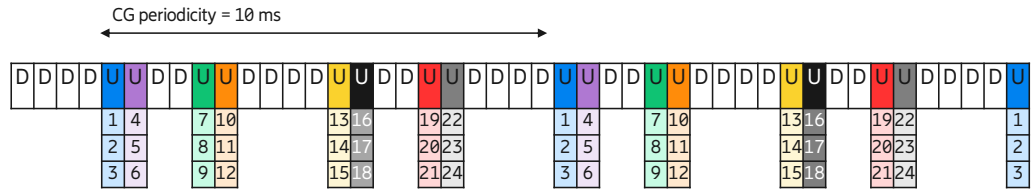


Figure 24 Uplink Configured Grant for URLLC Dedicated RAN capacity for CG periodicity=10 ms, CG/TTI=3 and TDD pattern DDDS1UUS2DUU

2.3.3.3.7 Analytical Model

The following model will provide the maximum number of users that can be served using configured grants for a given latency.

Definition of key parameters:

- Latency target, l [ms]
- Traffic periodicity, p_T [ms]
- CG periodicity, $P = \{2.5, 5, 10, 20\}$ ms
- t_{BS} = gNB processing delay
- t_{UE} = UE processing delay
- t_{slot} = slot duration
- n = number of retransmissions
- t_{retx} = HARQ retransmission delay
- $t_{SR-to-tx}$ = SR-to-transmit delay (excluding queuing)
- l' = budget for SR alignment and queuing delay
- U is the number of UL slots within l'
- s is the number CGs per TTI

The budget for SR alignment and queuing delay can be calculated as follows:



$$l' = l - t_{BS} - t_{UE} - t_{slot} - n \cdot t_{retx} [\text{ms}]$$

The CG periodicity P must be chosen so that $P \leq l'$

The number of slots within l' depends on the selected TDD pattern

- $U = \frac{P}{2.5 \text{ ms}}$ (DDDSU) 1 uplink slot every 2.5 ms
- $U = \frac{P}{5 \text{ ms}} \cdot 4$ (DDDS1UUDS2UU) 4 uplink slots every 5 ms

Finally, the number of devices is calculated as

$$C = U \times s$$

Besides the product constraints (e.g., number of CGs per TTI), the other main capacity limitation is bandwidth. Even though the packets sent using configured grant are small, to ensure their correct delivery MCS0 is used by default (see Sections 2.3.3.3.5 and 2.3.3.4). Thus, the required bandwidth is not negligible. Carrier bandwidth requirements are described in more detail in Section 2.3.3.4.

2.3.3.3.8 Results

Based on the analytical model described before and on the following assumptions:

- $t_{BS} + t_{UE} + t_{slot} = 2 \text{ ms}$,
- 0 application packets can arrive after the latency budget, and
- every packet is transmitted in a single uplink slot

the maximum number of devices according to the requested latency for different CG/TTI, configured grant periodicities and TDD patterns is calculated in Figure 25, Figure 26 and Figure 27.

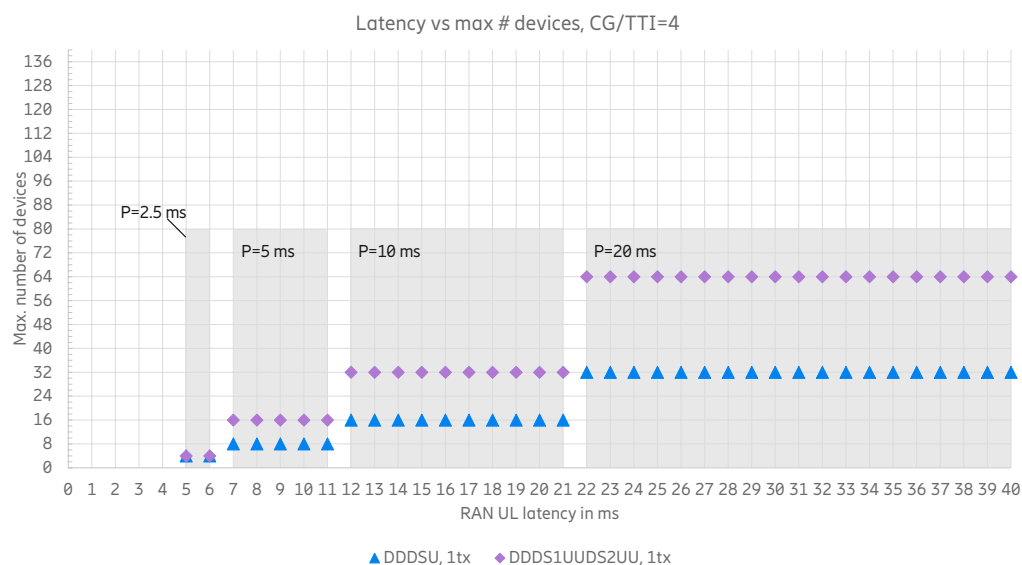


Figure 25 Maximum number of devices per cell for a CG/TTI=4

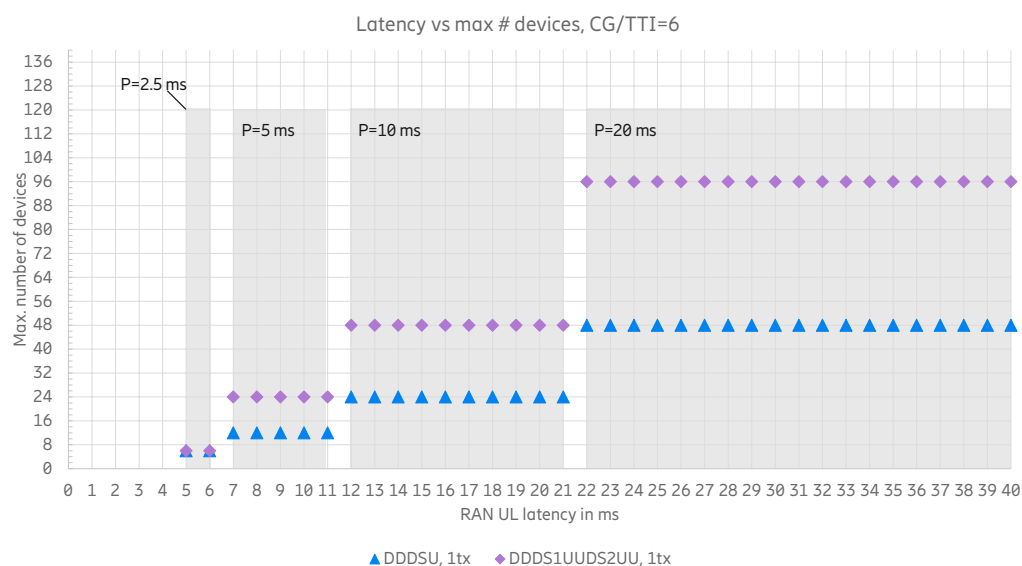


Figure 26 Maximum number of devices per cell for a CG/TTI=6

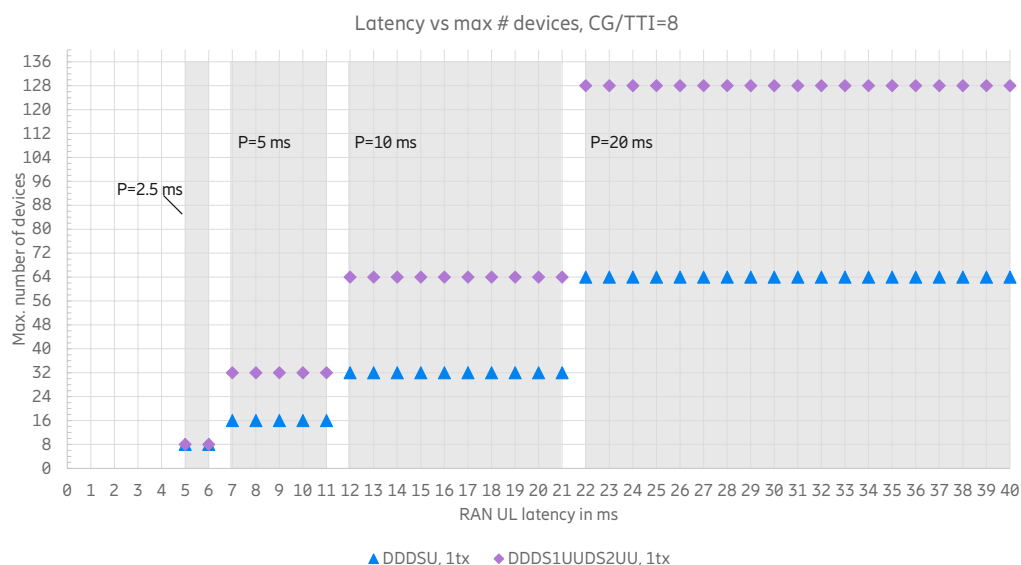


Figure 27 Maximum number of devices per cell for a CG/TTI=8

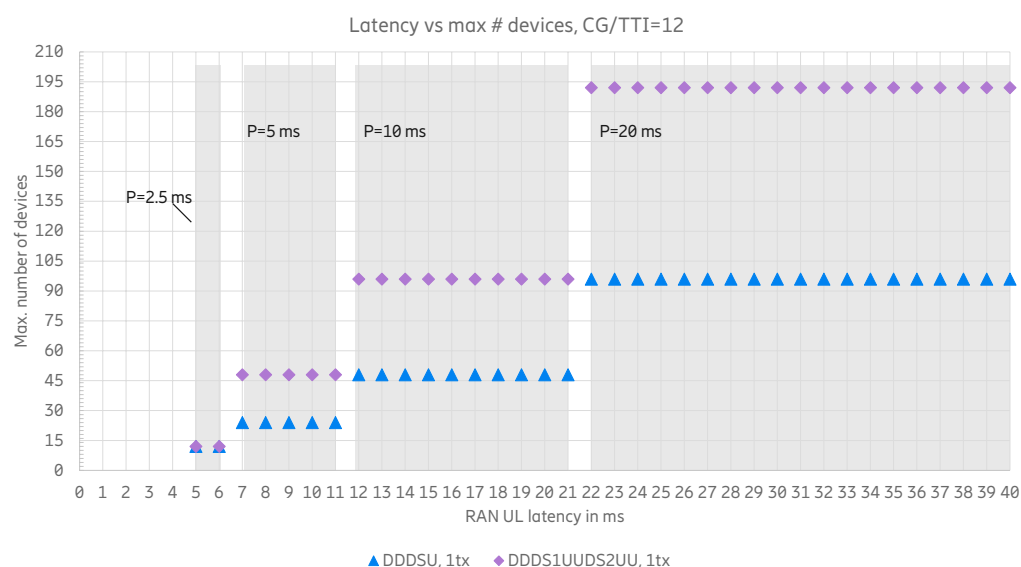


Figure 28 Maximum number of devices per cell for a CG/TTI=12

From these results, the following conclusions can be made:

- Short grant periodicity is required to achieve low latency. The shorter the grant periodicity, the lower the number of users supported.
- The shortest grant periodicity supported under these conditions is 2.5 ms which allows to achieve a minimum latency of 5 ms compared to the minimum latency in dynamic grant scheduling of 12 ms.
- Due to the distribution of uplink and downlink slots (i.e., there is a single pair of uplink slots with 2.5 ms periodicity in the DDDS1UUDS2UU TDD pattern), both patterns have the same capacity at a 2.5 ms periodicity.



- Because the DDDS1UUDS2UU pattern has twice as many uplink slots as the DDDSU pattern, it can support twice as many users for the same periodicity (except for $P=2.5$ ms).
- The maximum number of devices that can be supported can be increased increasing the CG/TTI, the uplink capacity of the TDD pattern or by targeting a higher latency.
- The same latency can be achieved by multiple configurations. For example, a specific latency can be achieved by using a long grant periodicity with a TDD pattern with high uplink capacity or by using a short grant periodicity with a TDD pattern with low uplink capacity.

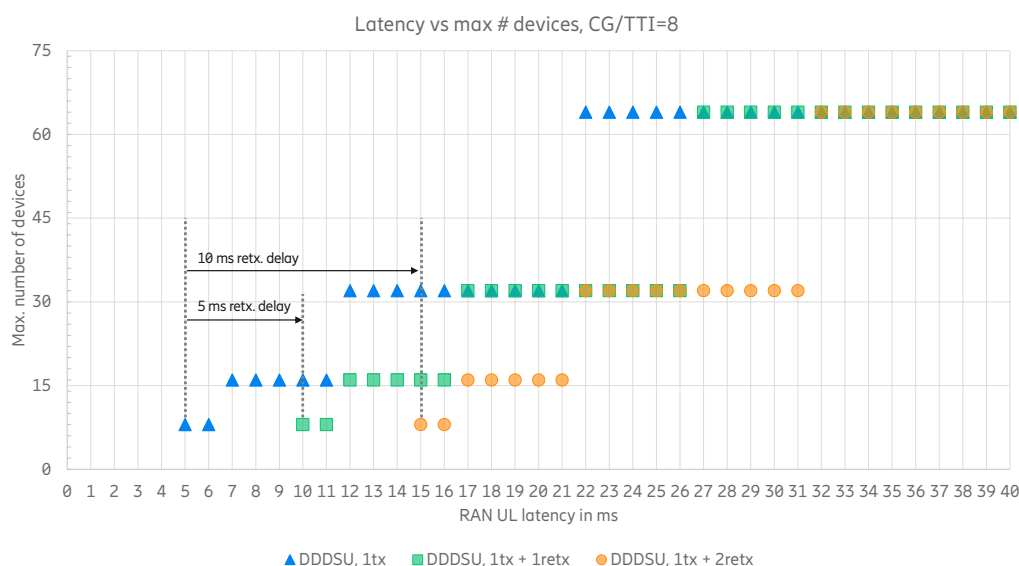


Figure 29 Maximum number of devices per cell for a CG/TTI=8 with 1 and 2 retransmissions

Similarly to dynamic grant scheduling, depending on the channel conditions retransmissions can be expected. In Figure 29, the effect of multiple retransmissions in the capacity curves can be seen. Assuming a 5 ms added delay for retransmissions (Section 2.3.3.1.5), the original graph from Figure 27 (without considering retransmissions) is shifted to the right by n times 5ms, where n is the number of retransmissions.

2.3.3.4 Carrier Bandwidth Dimensioning

Normally, it is not possible to choose how much bandwidth will be allocated to a carrier. Instead, planning is done around a given fixed carrier bandwidth. Therefore, it is important to understand what this carrier bandwidth means in terms of numbers of UEs with low-latency services. Therefore, the objective of this section is to show how to estimate the amount of bandwidth that it is required to serve UEs with low-latency services.

Spectrum radio resources required to schedule time critical low bitrate services depend on BLER target and channel conditions. There is a large variation in the required carrier bandwidth; evaluated for all modulation and coding schemes (MCS) in 3GPP Table 1 (TS 38.214).



In the case of dynamic grant scheduling and prescheduling, the MCS index used for transmission is selected dynamically by the gNB based on the radio conditions. The MCS is selected from a table (MCS Table 1, 2 or 3 in TS 38.214). If the channel conditions are poor, the UE will use a low MCS which ensures that the message will be properly decoded by the gNB at the expense of requiring more bandwidth. If the channel condition is good, a high MCS will be selected. The UE will send more information in the same bandwidth at the expense of having less protection in the packet and thus being more susceptible to interference or any type of issues in the radio channel.

In the case of configured grant, as link adaptation is not supported, the MCS is manually selected by a system constant. The default value is MCS0.

To estimate the required bandwidth, information about the packet size and the lowest expected Modulation and Coding Scheme (MCS) is required. As mentioned before, MCS can be variable depending on channel conditions. MCS information can be obtained through simulations or on-premises measurements.

The bandwidth B for a particular MCS in a single layer transmission can be calculated using the following equation

$$B(MCS) = s \frac{D_p}{T_p} \cdot \frac{T_s}{Q(MCS)R(MCS)} \frac{1}{(1 - OH)}$$

where s is the number of used configured grants or scheduling elements per TTI, D_p is the size in bits of the packet that will be transmitted and T_p is the time in seconds in which the packet must be transmitted. $Q(MCS)$ and $R(MCS)$ are the modulation order and coding rate, respectively, for a particular MCS, as indicated in the MCS Table 1 in TS 38.214. T_s is the OFDM symbol duration and OH represents the overhead for control channels.

Figure 30 shows the maximum required carrier bandwidth in MHz for different MCS indexes assuming a packet size of 128 bytes that must be transmitted in a single uplink slot. For the calculations of 4 SE/TTI (multiple users per slot), it was assumed that the same number of resources are allocated to each user.

Additionally, the following assumptions were done to calculate the results in Figure 30:

- $OH = 0.08$
- $T_s = \frac{10^{-3}}{14 \cdot 2}$, for numerology 1 sub carrier spacing 30 kHz
- $T_p = T_s$, the entire packet is transmitted in a single uplink slot

The calculations can be applied to scheduling using either dynamic grants or the Uplink Configured Grant for URLLC Dedicated RAN feature.

The results of Figure 30 show that high bandwidth is required at low MCS even for short packets.

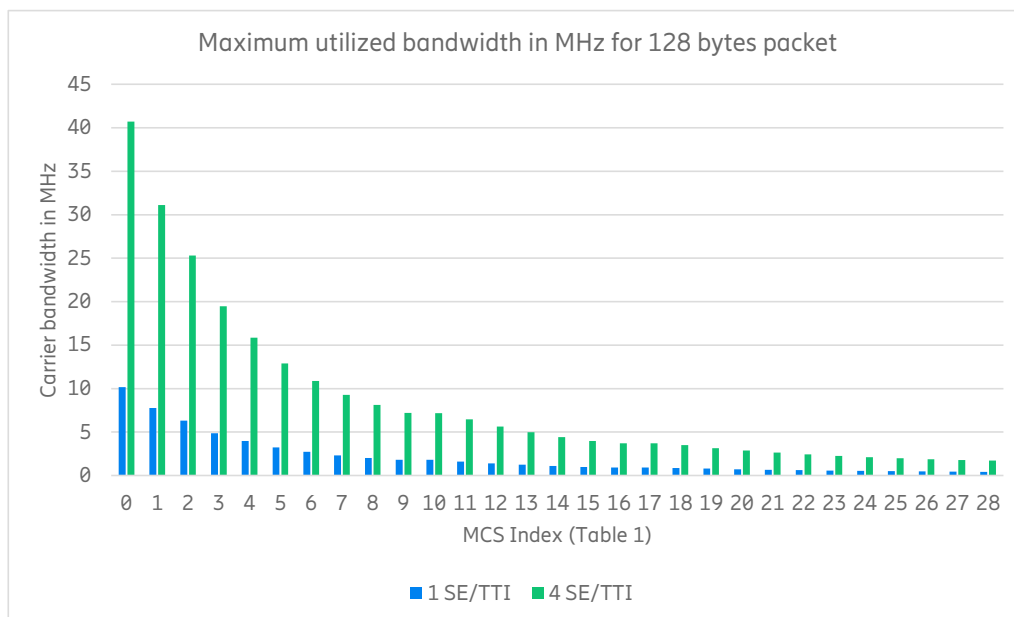


Figure 30 Maximum carrier bandwidth requirements for different MCS indexes and a 128 Bytes packet using 3GPP Table 1 (TS 38.214)

The calculations of the required bandwidth should be done assuming the worst MCS index expected in the factory.

Assuming an MCS 14 and packet size of 128 bytes, about 1 MHz is required to serve a single user per TTI, while approximately 5 MHz are required for a configuration with 4 users per TTI.

However, for MCS 0, which is the default value for packets sent using the Uplink Configured Grant for URLLC Dedicated RAN feature, the bandwidth requirements can go up to 10 MHz to send a single 128 bytes packet in a single uplink slot.

In conclusion, simulations or site measurements are necessary to discover the minimum MCS index that will be required. Based on this MCS index, the required bandwidth for the transmission of a single SE/TTI can be calculated. The actual number of SE/TTI that can be supported can be calculated by dividing the available bandwidth for TCC services by the bandwidth required by a single SE/TTI.

In the case of uplink configured grant, link adaptation is not currently available and thus the default MCS index is MCS0. For a 128 bytes packet approximately 10 MHz are required for a single SE/TTI. Thus, for 10 SE/TTI approximately 100 MHz of bandwidth are required. To avoid this large bandwidth consumption is recommended to check if the default MCS can be changed to a larger value (See Section 2.3.3.3.5 for more information on the selection of MCS for uplink configured grant).



2.3.3.5 Mixed Services Dimensioning

In a factory, a mix of time-critical and non-time-critical services with different requirements is expected. Thus, this needs to be taken into consideration during dimensioning.

2.3.3.5.1 Features to Differentiate Users and their Requirements

2.3.3.5.1.1 UE Grouping Framework:

This feature allows to differentiate UE handling for different groups of UEs (e.g., time critical and non-time critical UEs). Users can be differentiated by 5QI, ARP, S-NSSAI among others (See [UE Grouping Framework](#) for more information).

This feature can be used to select which users will be allowed to use the NR Service-Adaptive Prescheduling or the Uplink Configured Grant for URLLC Dedicated RAN features.

A current limitation is that the grouping is done on a UE basis. Therefore, if a single UE is used for both time-critical and non-time-critical services (e.g., UE is running a time critical service and operation and maintenance services), it will still be allocated to a single group. Thus, the configuration planned for that group may not fit both time-critical and non-time-critical services requirements.

2.3.3.5.1.2 NR Service-Adaptive Periodicity

Combined with the UE Grouping Framework this feature reserves access to SR periodicity pools to specific groups of users.

In this way, it is possible to reserve RRC SR periodicity pools with short SR periods for groups of UEs with low latency services.

2.3.3.5.1.3 NR Service-Adaptive Link Adaptation

Combined with the UE Grouping Framework this feature can set different initial grant sizes and target BLERs for different groups of UEs.

Being able to change the initial grant size is important in UEs used for time critical services so that the initial size grant matches the size of the transmitted packet and segmentation is avoided. However, the packet size required by non-time-critical services is usually smaller. By using the NR Service-Adaptive Link Adaptation feature, the SR grant size can be adjusted for both types of users independently. This optimizes resource consumption.

It is highly recommended to not mix different traffic types on the same UE. If this cannot be avoided and the same UE carries a mix of large and small packets with many small packets, it is recommended to adjust the SR grant size according to the small packet size to avoid resources being wasted.



If the SR grant size is larger than the actual packet size, the UE will add extra padding to match the SR grant size. This increases the utilization of PUSCH resources.

In a similar way, the target BLER of UEs with time-critical services needs to be adjusted according to reliability and latency requirements, while the BLER target from UEs non-time-critical services is usually higher to maximize throughput. Therefore, using this feature target BLERs can be assigned to UE groups according to their requirements.

2.3.3.5.1.4 NR Service-Adaptive Prescheduling

In high load conditions, users with basic priority prescheduling will be down prioritized compared to dynamic or configured grant users.

2.3.3.5.1.5 Uplink Configured Grant for URLLC Dedicated RAN

In case that uplink configured grant and dynamic grant users co-exist, the resource allocation for both types of users is controlled by the system constant `cgMaxResource`.

The system constant `maxNoOtaPuschSePerSlotPerCell` defines the maximum number of uplink SEs over the air that are allowed per slot and per cell. These scheduling elements are shared between configured and dynamic grant users. The `cgMaxResource` system constant defines the percentage of `maxNoOtaPuschSePerSlotPerCell` that will be allocated to configured grant users.

If a UE belongs to a group that requires uplink configured grant and there are not enough resources available, the UE will be scheduled using dynamic grant.

2.3.3.5.2 Carrier Bandwidth Considerations

Beyond the scheduling element capacity limitations on the baseband, another resource that must be shared between UEs with time-critical and non-time-critical services is bandwidth.

As discussed in Section 2.3.3.4, the bandwidth required by time-critical services depends on:

- Packet size
- SE/TTI
- Modulation and coding scheme index

In Figure 31, an example of the carrier bandwidth requirements for a factory with mixed services is shown. The factory has a 50 MHz carrier bandwidth available. Simulations indicate that the lowest MCS that is expected is 14 and 8 SE/TTI are sufficient for UEs with low-latency services. For a packet size of 128 bytes and the 8 SE/TTI, approximately 10 MHz are required for time critical



services. The rest of the bandwidth is available to account for overhead and other non-time critical services.

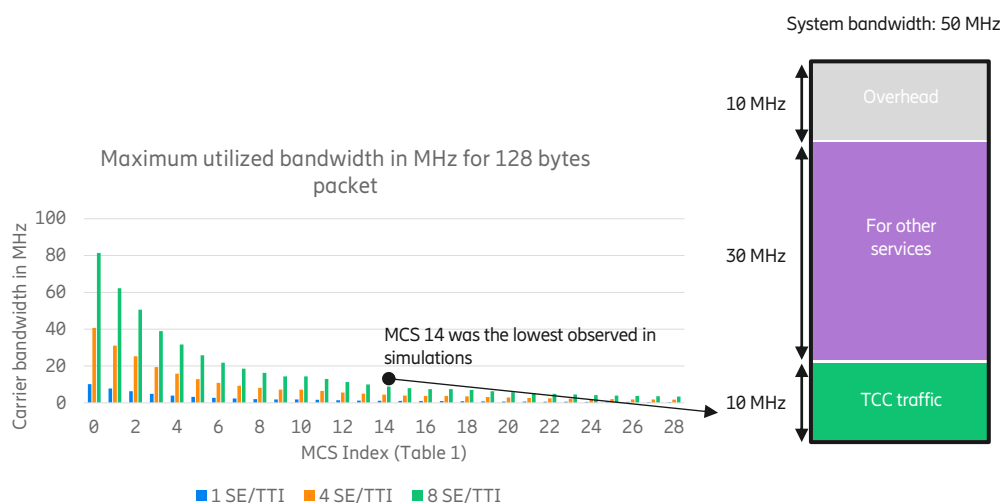


Figure 31 Example of bandwidth dimensioning for factory with mixed services

2.4 Radio Network Planning and Deployment Considerations

2.4.1 SINR considerations

When designing the radio network for time critical communication services it is important to note that designing for signal strength, cell coverage and throughput is not sufficient, but it is equally important to ensure good signal quality (SINR) throughout the network as well as sufficient resources in both signaling, e.g., PDCCH and processing per cell and baseband, as well as targeted applications.

Dimensioning of signaling and processing resources are addressed in Section 2.3.

Possible causes of low SINR may be:

- out-of-band emissions from macro on adjacent channels.
- EMF from machines on factory floor.
- in a multi cell deployment SINR is expected to deteriorate due to inter-cell interference and hence care need to be taken so sufficiently good SINR can be achieved in all areas serving traffic.

2.4.2 Radio Considerations for Deployments with Multiple Cells

The deployment of multiple cells in a factory is not recommended due to the possibility of intercell interference and handover issues. However, if it is required because of coverage or capacity requirements it must be carefully planned.



In the case that the cells reuse the same frequency carrier, intercell interference is possible in overlapping zones. As mentioned in the section before, interference has a negative effect on the SINR which affects the capacity of both data on control channels. Therefore, the overlapping area must be carefully planned:

- It is recommended to use firewalls and columns as blockers to control coverage area of the cells.
- In the case of buildings with high ceilings, it is recommended to lower the height of the radio dots to limit its coverage area.

Another issue with having multiple cells is that UEs moving from one cell to another one must perform a handover. A handover can be undesirable because of the added latency and service interruption that it implies. Therefore, the pathway of users with strict latency requirements such as automated guided vehicles (AGVs) should be preferably covered by the same logical cell.

2.4.2.1 NR Combined Radio (FAJ 121 5505)

The [NR Combined Radio](#) feature can be used to combine multiple radio units of the same type to form a single logical cell.

In the case of a factory environment, a single remote radio unit (RRU) or indoor radio unit (IRU) may not be sufficient to cover a large area.

Traditionally, each radio unit corresponds to a single cell. Therefore, having multiple radio units implies having multiple cells.

Having multiple cells may not be desirable due to possible intercell interference and handover issues mentioned before.

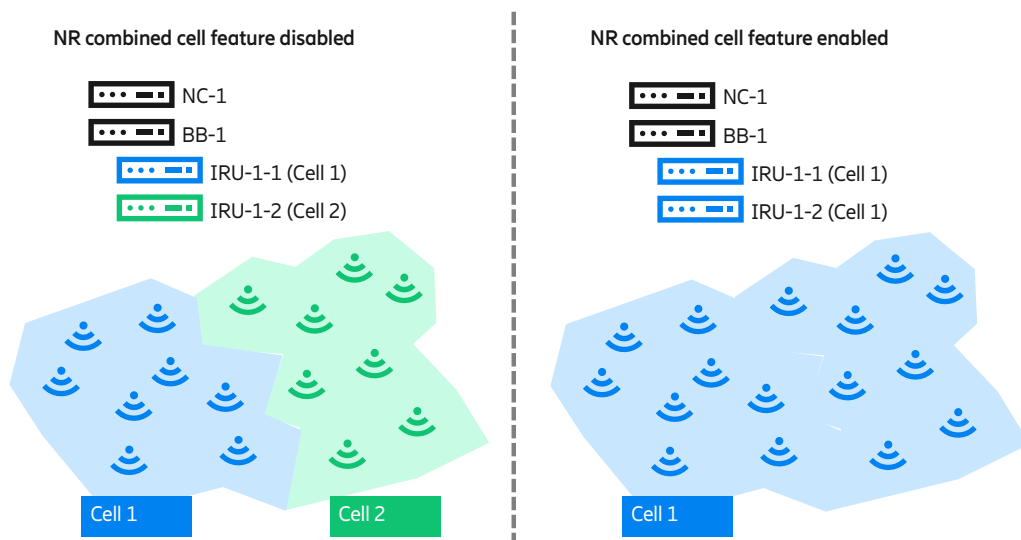


Figure 32 NR combined radio feature

To avoid the increased interference and number of handovers, in a multiple cell scenario the NR Combined Radio feature can be used, as shown in Figure 32, to combined multiple radios in a single logical cell.



For more information, see [NR Combined Radio on RDS](#) and [NR Combined Radio](#).

2.4.2.2 Cell Grouping

The cell grouping feature allows for the grouping of a certain number of radio dots into a single cell. The dot grouping feature allows a subset of dots to carry a subset of carriers.

If multiple cells must be deployed, due to capacity or coverage reasons, the cell grouping feature can be used to reduce the number of required Indoor Radio Units (IRUs).

Normally, all dots connected to a single IRU form part of the same cell. However, by using the Cell grouping feature it is possible to split the dots into 2 independent cells.

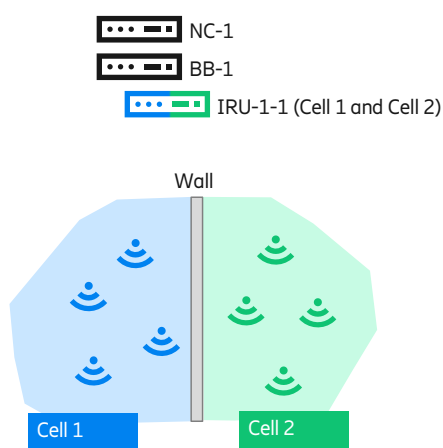


Figure 33 Cell grouping example

Figure 33 shows an example of cell grouping. In this case, the factory floor is split in two parts by a wall. Each half of the factory floor only requires 4 dots for coverage. Because the wall provides isolation, 2 cells reusing the same carrier can be deployed. Normally, this would require 2 IRUs. However, because only 4 dots are required to cover each half of the factory floor, the cell grouping feature can be used to group the 8 dots of a single IRU into 2 cells with 4 dots each.

For more information, see [Cell Grouping on IRU 88xx](#).

2.5 QoS Considerations

Assigning appropriate QoS identifiers to services with low latency requirements it is important so that the network can provide the appropriate treatment to this type of traffic specially in the presence of traffic from non-low-latency services.



The [QoS Framework](#) feature makes it possible to establish QoS flows that provide optimized network treatment for traffic flows with low-latency requirements, in addition to the default QoS flow used for MBB traffic.

A dedicated bearer is needed to get a higher priority level than the default bearer. Therefore, it is suggested to configure a dedicated flow for the low latency services. With a dedicated bearer, it is possible to configure different scheduling profiles that are optimized for time-critical service. The default bearer is typically mapped with MBB traffic, so it is recommended to use low-priority and similar scheduling profiles for services that do not require low latency.

Multiple QoS flows per PDU session can be used to isolate the traffic from low-latency services from the MBB traffic. An example of a DU5qiTable configuration is shown in Table 8.

Attribute	DU5qi	DU5qi
dU5qiId	1	2
priorityLevel	90	68
profile5qi	9	80

Table 8 Example of GNBFunction.DU5qiTable.DU5qi Configuration

According to the rule of the PDU session, one specific UPF must be used to terminate the GTP-U tunnel of the PDU session. All QoS flows in the PDU session must be served by such UPF.

For more information on the QoS configuration for networks with time critical services, see the [Critical IoT Solution Guideline](#).

For information about the end-to-end perspective of QoS including the Transport domain, see [E2E QoS Guideline](#).



3 Network Elements and RAN Features List

3.1 RAN Feature List

For detailed capacity information on the NR RAN software, see the following document:

- [Supported Capacity and Configurations for NR RAN](#)

The features required to configure dynamic grant scheduling for time critical low-rate services are listed in Table 9.

Feature	Description	Value Package
NR Service-Adaptive SR Periodicity (FAJ 121 5595)	By introducing the feature NR Service-Adaptive SR Periodicity, it is possible to configure different SR periodicity per UE group and to configure the number of UEs allocated per SR periodicity.	Performance Boost (FAJ 801 4021)
LTE-NR FDD Spectrum Sharing with NR Standalone (FAJ 121 5257)	The feature allows the dynamic resource sharing of the same spectrum for load adaptation between an LTE cell and an NR cell deployed in an NR Standalone (SA) configuration. This allows the simultaneous deployment of LTE and NR-SA services in the same geographical area.	LTE-NR FDD Ericsson Spectrum Sharing (FAJ 801 4012)
NR Service-Adaptive Link Adaptation (FAJ 121 5513)	This feature enables a configuration framework for service-adaptive link adaptation in Low-Band and Mid-Band cells.	Performance Boost (FAJ 801 4021)
QoS Controlled SR scheduling (FAJ 121 5157)	The feature enhancement improves QoS handling in uplink scheduling in high load by optimizing the	NR Base Package (FAJ 801 4002)



	scheduling priority of scheduling requests.	
--	---	--

Table 9 RAN features to reduce latency in dynamic grant scheduling
Features required for NR prescheduling are listed in Table 10.

Feature	Description	Value Package
NR Service-Adaptive Prescheduling (FAJ 121 5324)	The NR Service-Adaptive Prescheduling feature introduces the possibility to configure different prescheduling modes for each predefined UE group.	Performance Boost (FAJ 801 4021)

Table 10 RAN features to enable NR prescheduling
Features required for Uplink Configured Grant scheduling are listed in Table 11.

Feature	Description	Value Package
Uplink configured grant for URLLC Dedicated RAN (FAJ 121 5408)	The feature significantly reduces latency by avoiding the delay associated with the scheduling request and the scheduling grant.	Critical IoT for Dedicated RAN Base Package (FAJ 801 1701)

Table 11 RAN features to enable configured uplink grant scheduling
Features required to differentiate users (e.g., time critical from MBB users) are listed in Table 12.

Feature	Description	Value Package
UE Grouping Framework (FAJ 121 5314)	The UE Grouping Framework introduces a solution to handle the demand for the differentiation of services, functions, and feature behavior.	NR Base Package (FAJ 801 4002)
NR Radio Resource Partitioning (FAJ 121 5337)	The feature ensures that resource partitions get prioritized access to PRBs corresponding to the resource shares configured in the	Advanced RAN Slicing (FAJ 801 4019)



	resource allocation policy.	
--	-----------------------------	--

Table 12 RAN features to enable user differentiation

Features required in deployments with multiple cells are listed in Table 13

Feature	Description	Value Package
NR Combined Radio (FAJ 121 5505)	The feature enables the configuration of multiple radio units acting as one radio unit, creating one logical cell.	NR Base Package (FAJ 801 4002)
NR Advanced Multi-layer Coordination (FAJ 121 5584)	The feature ensures that traffic load is considered in traffic steering when selecting best cell set for individual UE.	Intelligent Traffic Management and Mobility (FAJ 801 4022)

Table 13 RAN features to support deployments with multiple cells

3.2 Radio and Transport Hardware

For the full component list of hardware used in Ericsson Private 5G, please see [Technical Product Description](#) in Ericsson Portfolio and Commerce.



4 Network Design – Factory Example

4.1 Example 1 – Factory 1 with PROFINET Connections

4.1.1 Overview

Example network targets an Ericsson Private 5G factory deployment delivering time (latency) critical services.

4.1.2 Service Requirements

The example network is considering a factory environment composing of both latency-critical and non-latency-critical services.

The Latency-critical services (Time Critical) is considering a fleet size of 50 connected AGV (Automatic Guide Vehicles) for automated indoor transportation of manufacturing parts. Fleet AGVs referenced in example are utilizing "PROFINET" UEs to communicate with centralized functions for vehicle control and monitoring.

For non-latency-critical services the Large Type AGV is equipped with two additional UEs for communication with the onboard touchscreen (HMI).

The fleet are, comprised of two vehicle variants as described below with respective traffic profiles. Uplink and Downlink is to be considered as measurement-average over time.

4.1.2.1 AGV Type Large

Number of Vehicles – 25

Number of UE - 3

UE1: PROFINET control traffic (UDP):

- Uplink: 330 bytes / packet, one packet every 128ms
 - Assumption: 500 bytes with overhead
- Downlink: 330 bytes / packet, one packet every 128 ms
- Watchdog timer 512ms (loss of 4 packets allowed)
- F-Monitoring time/Time out value: not used.

UE2 and UE3 connected to a touch screen (HMI) with TCP traffic generated upon interaction:

- Uplink: 362 kbps
- Downlink: 285 kbps



4.1.2.2 AGV Type Small

Number of Vehicles – 25

Number of UE - 1

UE1: PROFINET control traffic (UDP):

- Uplink: 330 bytes / packet, one packet every 128ms
 - Assumption: 500 bytes with overhead
- Downlink: 330 bytes / packet, one packet every 128 ms
- Watchdog timer 512 ms
- F-Monitoring time/Time out value: not used.

4.1.3 Physical Site Design

Considered for later releases.

4.1.4 Transport Design

Considered for later releases.

4.1.5 Radio Network Design

Figure 34 describes the radio planning done at the factory. The factory has approximately 80 000 m² (300x275 m).

100 MHz of available bandwidth in 3700 – 3800 MHz are split in 2 carriers (i.e., F-1 and F-2) of 50 MHz each. The selected TDD pattern is DDDSU. A total of 4 cells are deployed. Cells 1 and 4 reuse the carrier F-1 and cells 2 and 3 reuse carrier F-2. The area covered by cells 1 and 2 is separated by a firewall from the area covered by cells 3 and 4. Therefore, interference between cells 2 and 3 or between cells 1 and 4 is not expected.

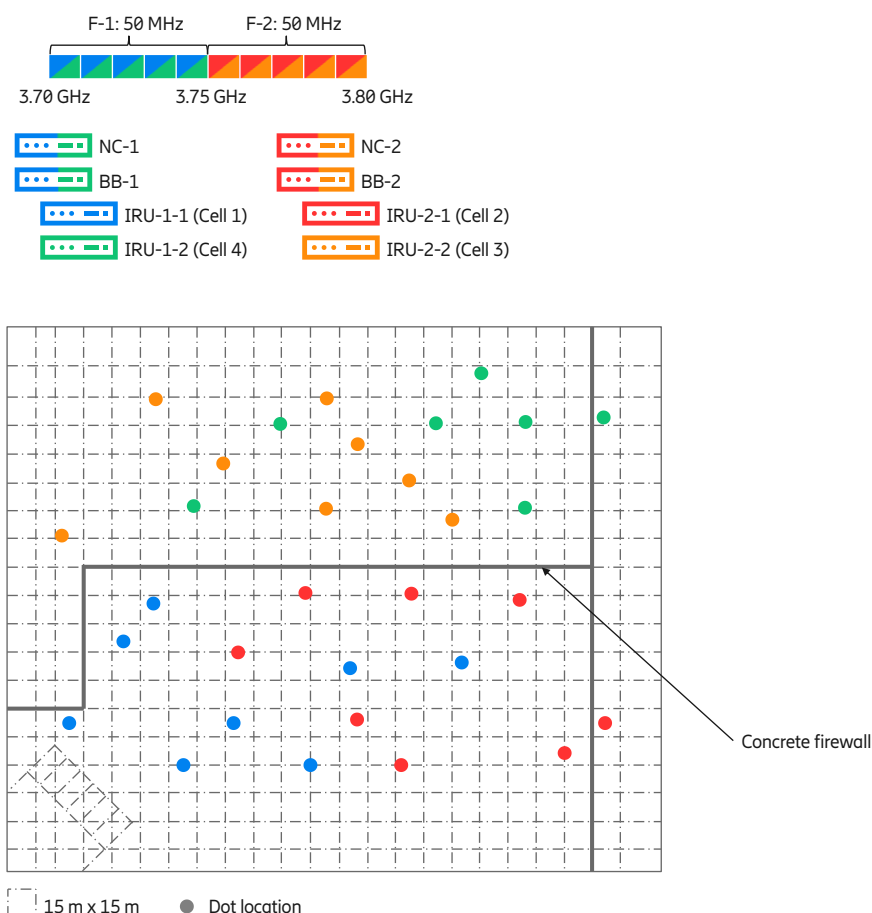


Figure 34 Factory layout and network setup

As shown in Figure 35, to provide redundancy, there is a large overlap between cells 1 and 2 and between cells 3 and 4. In case of cell failure, the remaining cell must be capable of serving the UEs providing time critical services.

To provide baseband and IRU redundancy, the overlapping cells are connected to different basebands and IRUs.

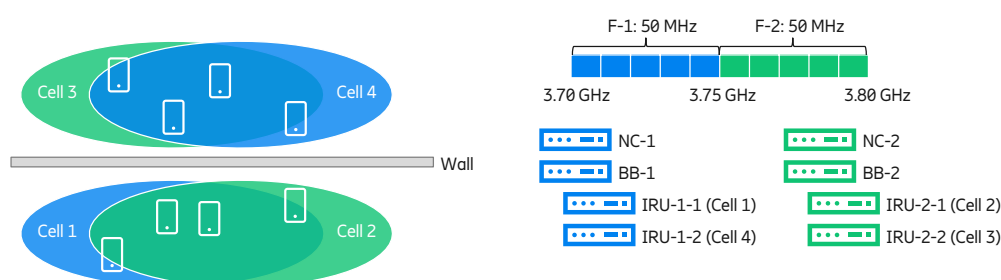


Figure 35 Factory radio planning with redundant cells



4.1.5.1 RAN Latency Dimensioning

Figure 36 shows an estimation of the required bandwidth. MCS14 was the lowest MCS index observed in the simulations done for this scenario. From this, the following conclusions can be made:

- More than 300 MHz are required for MCS0 and SE/TTI=8.
- 35 MHz are required for MCS14 and SE/TTI=8.
- MCS0 is the default value for the Uplink Configured Grant for URLLC Dedicate Networks feature. As only 50 MHz are available, the Uplink Configured Grant for URLLC Dedicate Networks feature cannot be used unless a different MCS index is manually configured.
- Using a larger SE/TTI will require more bandwidth for time-critical services and would leave no bandwidth for any additional services.

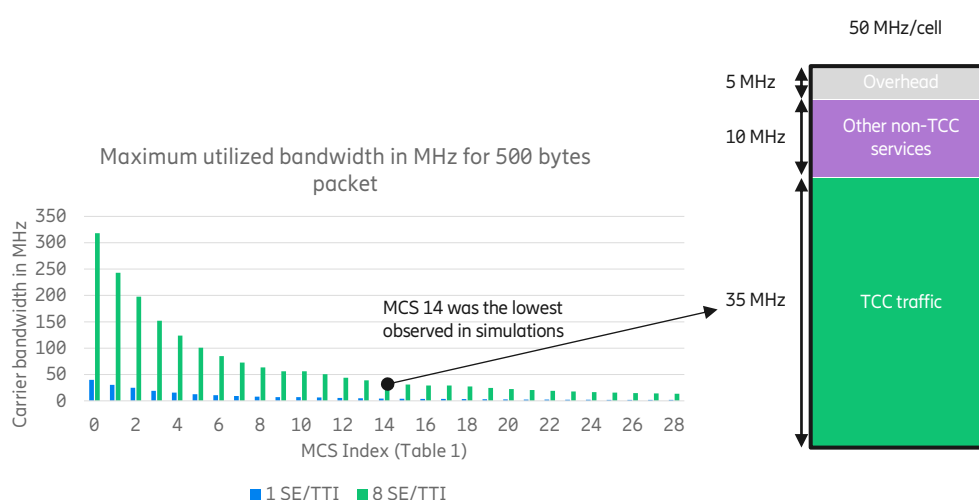


Figure 36 Carrier bandwidth estimation

Based on these conclusions, latency calculations are done in Figure 37, Figure 38 and Figure 39 for the case that uplink configured grant, dynamic grant or prescheduling is used.

To account for reliability, a single retransmission is considered during the latency calculations. Therefore, one SE/TTI is reserved for retransmission. Calculations were done considering SE/TTI=7.

From this, the following conclusions can be made:

- A capacity of 56 users can be achieved with the 3 scheduling methods. However, the uplink RAN latency for uplink configured grant is 27 ms, the latency for prescheduling 27.5 ms and for dynamic grant scheduling is 32 ms.
- Uplink configured grant has the lowest uplink RAN latency but it requires UE support and link adaptation is not yet supported. The default MCS must be changed to MCS14
- Prescheduling has a similar uplink RAN latency as uplink configured grant. However, UEs with non-low-latency services are expected in the network.



UEs using prescheduling with basic priority can be down prioritized over UEs with non-low-latency services.

- Dynamic grant scheduling has the largest uplink RAN latency of all scheduling methods, 32 ms, however it does not require a UE with special capabilities, and it does not have the down prioritization problems of prescheduling.

Based on the observations listed above, dynamic grant scheduling is selected as scheduling method.

In order to avoid segmentation, the srGrantSize MoM parameters in the [NR Service-Adaptive Link Adaptation](#) (Section 2.3.3.1.5.1) must be set to $500 \times 8 = 4000$ bits for the group of UEs with low-latency services.

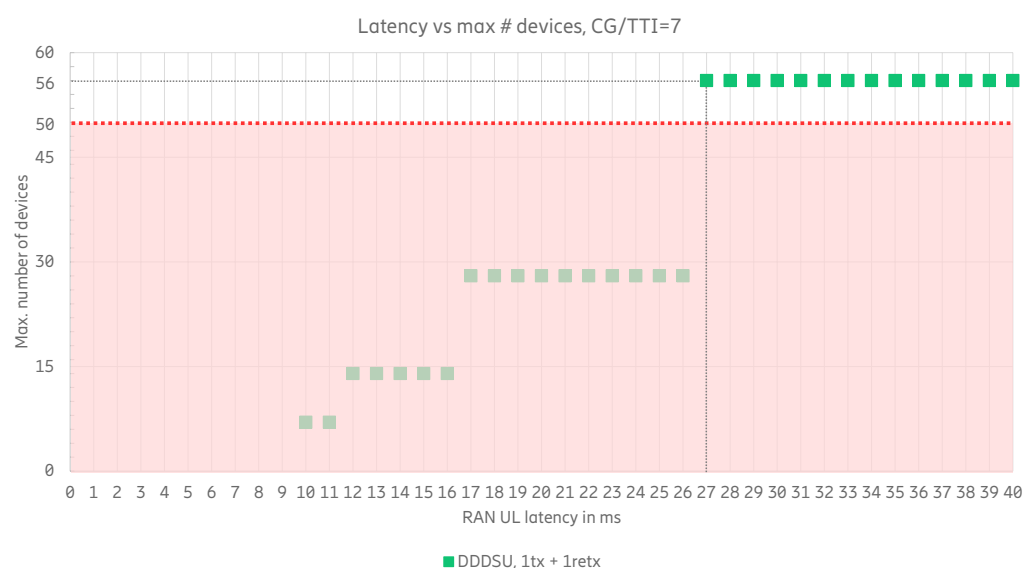


Figure 37 Capacity estimation using the Uplink Configured Grant for URLLC Dedicate Networks feature and CG/TTI=7

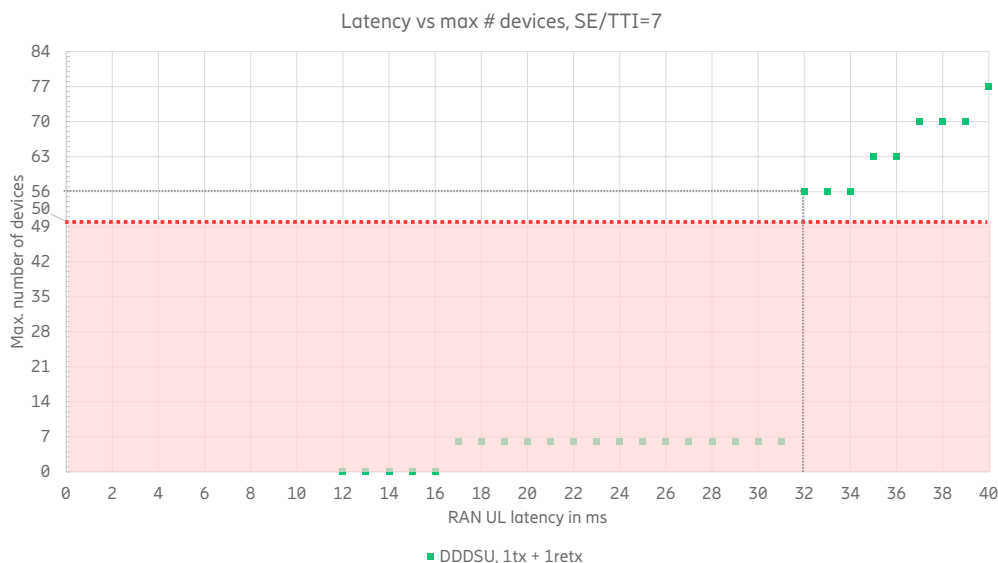


Figure 38 Capacity estimation using dynamic grant scheduling (default RRC SR periodicity pools - Table 4) and SE/TTI=7

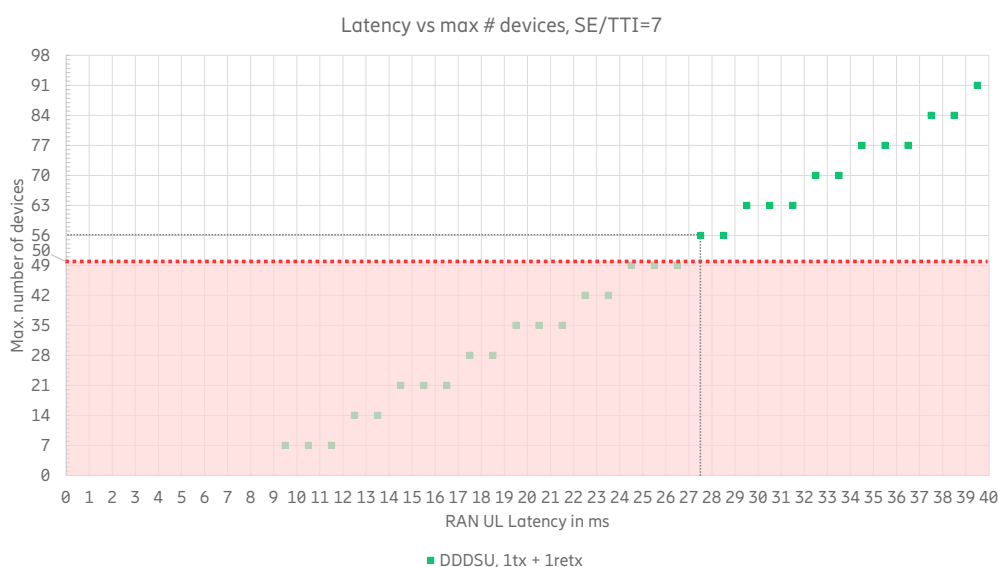


Figure 39 Capacity estimation using NR service-adaptive prescheduling feature and SE/TTI=7

4.1.6 UE / Device

In total 50 PROFINET UEs connected

- Chipset used in example design is Qualcomm QC X55 (PROFINET).
- For detailed specifications on Qualcomm QC X55 contact Ericsson [DSI](#) organization.



4.1.7 Observability

Considered for later releases.

4.1.8 Positioning

See the [Indoor Positioning Guideline](#) for more information.

4.2 Example 2 – Factory 2 with PROFINET Connections

4.2.1 Overview

Example network targets an Ericsson Private 5G factory deployment delivering time (latency) critical services.

4.2.2 Service Requirements

The example network considers a factory environment composed of only latency-critical services.

The latency-critical services (Time Critical) is considering a group of 80 UEs connected to a PROFINET device that uses:

- 128 ms cycle time (end to end one way latency)
- 768 ms Watch dog timer (survival time)
- 200 Byte packet size, application layer
 - Assumption: 400 bytes with overhead

4.2.3 Radio Network Design

100 MHz of available bandwidth in 3700 – 3800 MHz are split in 2 carriers (i.e., F-1 and F-2) of 50 MHz each. The selected TDD pattern is DDDSU. A total of 2 cells are deployed. Cell 1 use the carrier F-1 and cell 2 the carrier F-2.

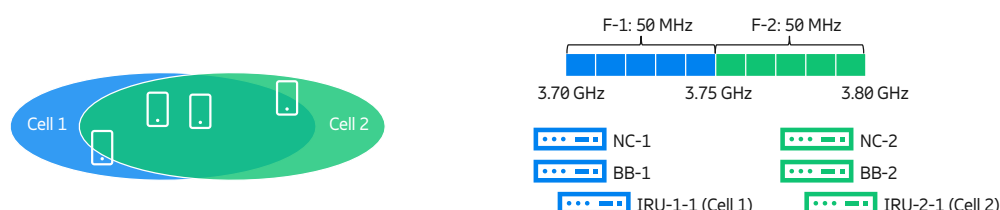


Figure 40 Factory radio planning with redundant cells

As show in Figure 40, to provide redundancy, there is a large overlap between cells 1 and 2. In case of cell failure, the remaining cell must be capable of serving the UEs providing time critical services. To provide baseband and IRU redundancy, the overlapping cells are connected to different basebands and IRUs.



4.2.3.1 RAN Latency Dimensioning

Figure 41 shows an estimation of the required bandwidth. MCS14 was the lowest MCS index observed in the simulations done for this scenario. From this, the following conclusions can be made:

- More than 350 MHz are required for MCS0 and SE/TTI=12.
- 42 MHz are required for MCS14 and SE/TTI=12.
- MCS0 is the default value for the Uplink Configured Grant for URLLC Dedicate Networks feature. As only 50 MHz are available, the Uplink Configured Grant for URLLC Dedicate Networks feature cannot be used unless a different MCS index is manually configured.
- Using a larger SE/TTI will require more bandwidth than what is available per cell.

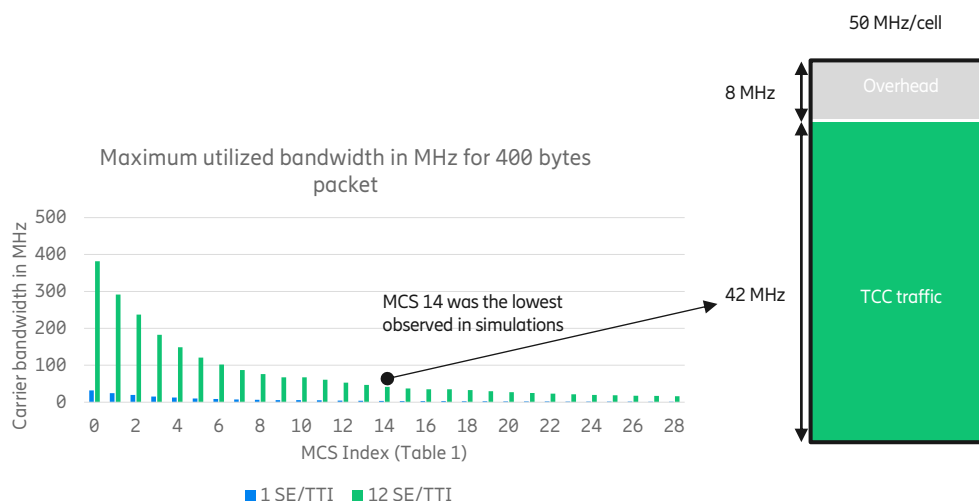


Figure 41 Carrier bandwidth estimation

Based on these conclusions, latency calculations are done in Figure 42, Figure 43, Figure 44 and Figure 45 for the case that uplink configured grant, dynamic grant or prescheduling is used.

To account for reliability, a single retransmission is considered during the latency calculations. Therefore, one SE/TTI is reserved for retransmission. Calculations were done considering SE/TTI=11.

From this, the following conclusions can be made:

- A capacity of 88 users can be achieved with the 3 scheduling methods. However, the uplink RAN latency for uplink configured grant is 27 ms, the latency for prescheduling 27.5 ms and for dynamic grant scheduling is 32 ms.
- Uplink configured grant has the lowest uplink RAN latency but it requires UE support and link adaptation is not yet supported. The default MCS must be changed to MCS14
- Prescheduling has a similar uplink RAN latency as uplink configured grant.



- Dynamic grant scheduling has the largest uplink RAN latency of all scheduling methods, 32 ms, however it does not require a UE with special capabilities.
 - Changing the default RRC SR periodicity pool (Table 4) to a customized RRC SR periodicity pool with more SR resources for low latency users (Table 6) did not improve the latency for the target of 80 users. In both cases 32 ms of RAN uplink latency are required for a capacity of 88 users.
 - Nevertheless, comparing the Region 2 of Figure 43 and Figure 44, shows that by using the customized RRC SR periodicity pool there is a user capacity increase. Still, a larger CG/TTI capacity or a TDD pattern with more uplink capacity are necessary so that this increase is large enough to have an impact in this example.

Based on the observations listed above and specifically considering that only low-latency traffic is expected in the network (i.e., no mix of non-low- and low-latency traffic), NR service-adaptive prescheduling is selected as scheduling method.

In order to avoid segmentation, the preschedulingDataSize MoM parameters in the [NR Service-Adaptive Prescheduling](#) (Section 2.3.3.2) must be set to 400 bytes for the group of UEs with low-latency services.

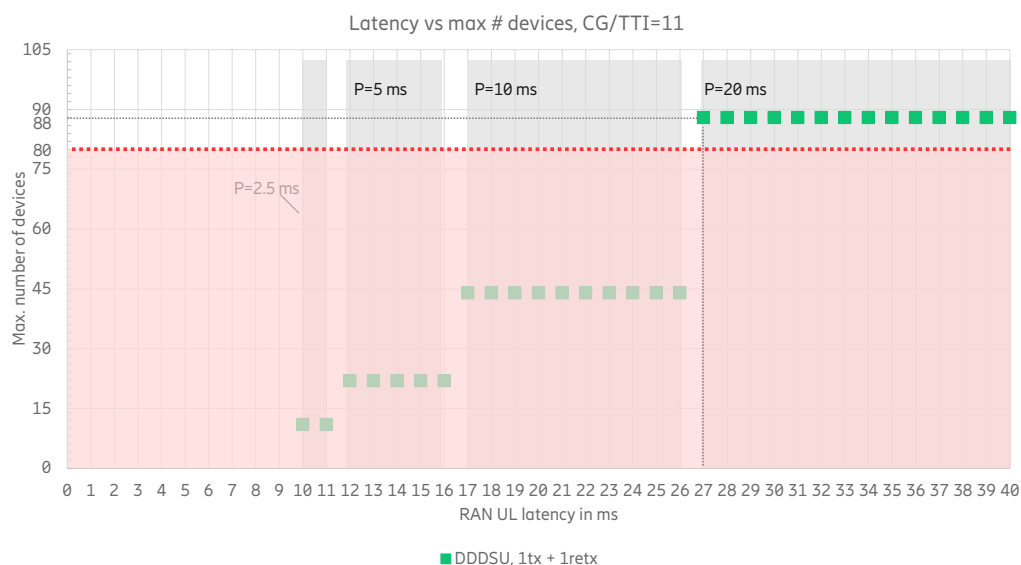


Figure 42 Capacity estimation using uplink configured grant feature and CG/TTI=11

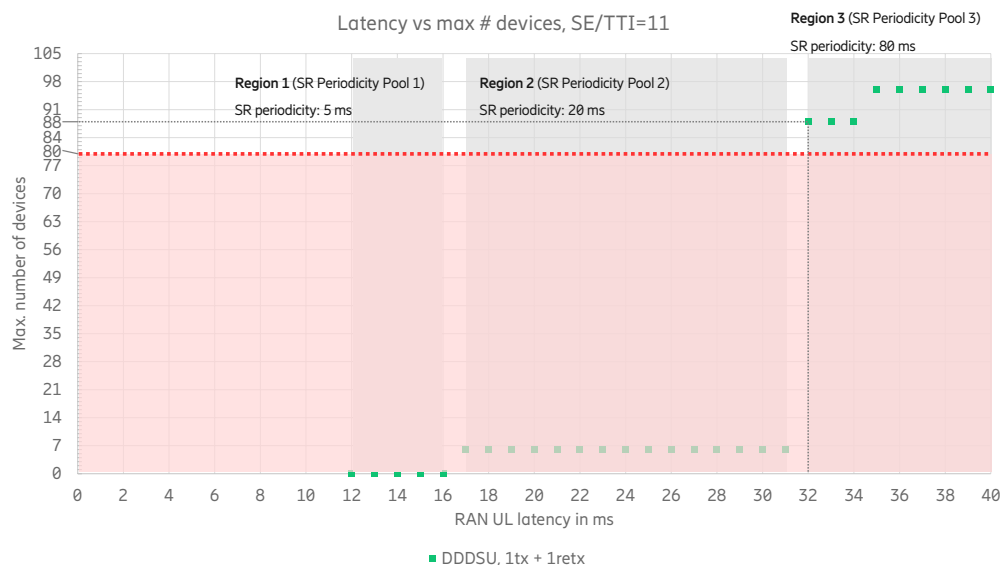


Figure 43 Capacity estimation using dynamic grant scheduling (default RRC SR periodicity pools - Table 4) and SE/TTI=11

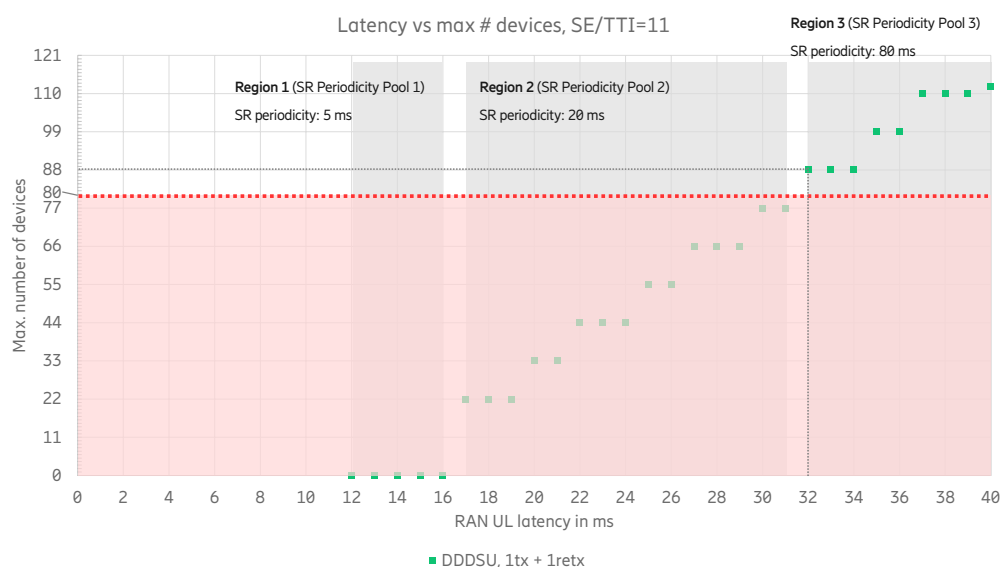


Figure 44 Capacity estimation using dynamic grant scheduling (customized RRC SR periodicity pools as defined in Table 6) and SE/TTI=11

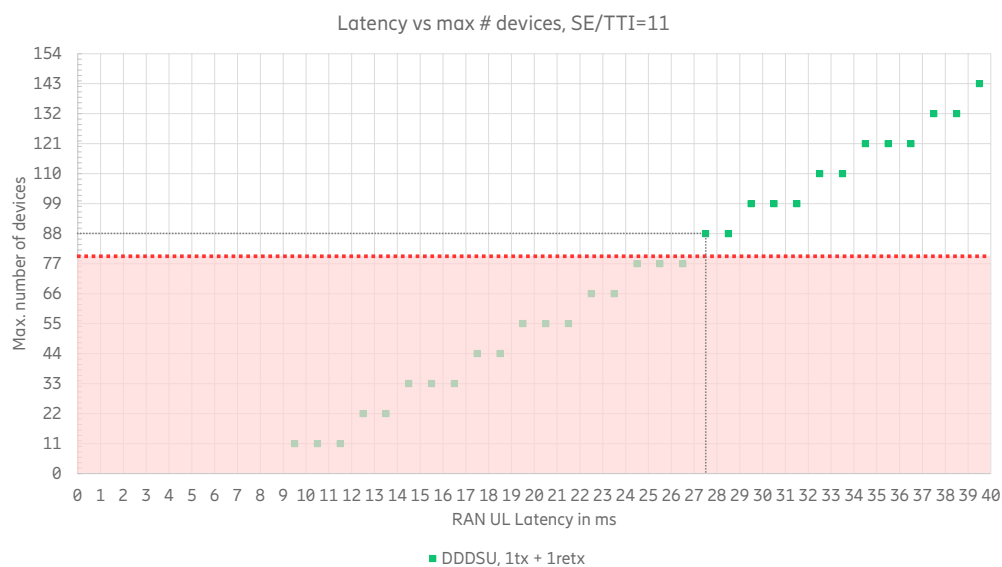


Figure 45 Capacity estimation using NR service-adaptive prescheduling feature and SE/TTI=11

5 Observability

Consider in future releases.

6 Troubleshooting & Best Practices

Consider in future releases.



Solution Guideline Change History

This section summarizes the major changes introduced in each document release.

Table 14 Main Solution Guideline Changes per Release

Release	Main Changes
23.Q4.0	<i>First version</i>